

TECHNICAL MANUAL

MIR 9000e

MULTIGAS EMISSION
& PROCESS MONITOR

- APRIL 2020 -



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CHAPTER 1

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GENERAL – CHARACTERISTICS



Figure 1–1 – MIR 9000e analyzer presentation.

GENERAL**PRESENTATION**

The MIR 9000e analyzer is designed for the continuous measurement of 1 to 7 compounds in a dry gaseous sample. It uses non-dispersive infrared (NDIR) spectroscopy. This method, based on a completely physical principle, allows for a specific measurement of each compound.

It can also measure oxygen via either a ZrO₂ or a Paramagnetic cell (optional).

The gas concentration is measured on a dry, clean, and non-explosive gas sample.

The MIR 9000e has many advantages, thanks to the latest electronic technology, and requires only limited maintenance.

The measurements are given by:

- A color touch screen mounted on the front panel
- A digital output
- Optional analog outputs

The analyzer can be used for various applications, and in particular:

- Combustion monitoring
- Monitoring of combustion installation exhaust
- Monitoring of household waste incineration plants flue gas emissions
- Monitoring of industrial waste incineration plants flue gas emissions
- Process monitoring
- Gas analysis in the chemical industry, oil industry, etc.

For each application, the sampling and the sample conditioning will need a specific study.

DESCRIPTION

Front panel

The front panel includes the following elements:

- StatusLight™ general power button (color changing).
- Backlit color touch screen:
 - Resolution of 800 x 480 pixels, 7"
 - Displays the measurement values in the chosen unit
 - Provides the main Human Machine Interface
- USB port



Figure 1-2 – Color touch screen

Rear panel

The sample inlet and outlet and the electrical connectors are located on the rear panel of the MIR 9000e. Refer to Figure 1–3.

Sample inlet and outlet

- The inlet for the sample to be analyzed (1) consists of a polypropylene fitting for 4/6 mm Teflon tube.
- The sample outlet (2) for the possible connection to a venting circuit consists of a polypropylene fitting for 4/6 mm Teflon tube.

The gas pressure at analyzer inlet and outlet must be close to atmospheric pressure.

Electrical equipments and connections

- A power supply three-contact socket (10) for mains connection via a standard cord comprising :
 - An EMC/EMI power supply filter,
 - An ON/OFF switch (11),
 - A 3.15 A / 250 V general fuse, 5x20 mm size, time delay type (12).
- One Ethernet port (3) and two USB ports (4),
- One connection for the 24V power supply of the Input / Output Module option (5),
- One 10-point BL connection to connect 1 to 5 external calibration solenoid valve(s) (6):

SOLENOID VALVES									
SV1 (Zero SV)		SV2 (Span SV)		SV3 (Span 1 SV)		SV4 (Span 2 SV)		SV5 (Probe SV)	
+24V	GND / open	+24V	GND / open	+24V	GND / open	+24V	GND / open	+24V	GND / open

- One 2-point BL auxiliary connection to connect a 24V external pump or an additional external solenoid valve (7),

SAMPLE PUMP or AUX	
+24V	GND / open

- One 4-point BL connection to connect an optional NOx converter oven (8),

NOx CONV.			
PT100	PT100	GND / open	+24V

- One 3-point BL connection to connect the analog signal of an external O2-probe (9), either voltage or current type (not both at the same time, because this is a single input):

O2 PROBE		
0-10V	4-20 mA	GND

- A 24V-2A fuse (13), to protect the (5), (6) and (7) outputs.

NOTE : The analyzer is equipped with a power supply compatible with a 100-240V voltage/50-60 Hz frequency mains.



(1) sample inlet, (2) sample outlet, (3) Ethernet port, (4) USB port, (5) 24V power supply connection of the Input / Output Module option, (6) 10-point BL connection to connect 1 to 5 external calibration solenoid valve(s), (7) 2-point BL auxiliary connection to connect a 24V external pump or an additional external solenoid valve, (8) 4-point BL connection to connect a dedicated NOx converter oven, (9) 3-point BL connection to connect the analog signal of an external O2-probe, either voltage or current type (not both at the same time, because this is a single input), (10) three-contact socket for mains connection via a standard cord, (11) ON/OFF switch, (12) 3,15 A / 250 V general fuse, 5x20 mm size, time delay type, (13) 24V-2A fuse to protect the (5), (6) and (7) outputs, (14) extractor fan.

Figure 1-3 – MIR 9000e rear panel

Internal view

WARNING: Make sure that the analyzer mains power cord is not connected.

Loosen the 5 screws holding the top cover, then remove the top cover to access the internal elements. The internal elements include the followings (Figure 1–4):

- Sample inlet (1)
- Sample inlet dust filter (2)
- Chamber pressure sensor (3)
- Pump (4)
- Pump pressure sensor (5)
- Mains filter (6)
- 24V DC power supply with universal input voltage 100~250VAC 50/60 Hz (7)
- O₂ probe (8)
- Measurement enclosure (9) consisting of :
 - Motor, IR source and correlation wheel assembly,
 - Optical bench assembly.
- Interconnection board (10) for interfacing the Power-supply, Arm20, Controller, and Measurement boards
- Power supply board (11)
- Arm20 board (12)
- Controller board (13)
- Measurement board (14)

The gas to be analyzed is drawn in by the pump (4) located downstream the fluid circuit. It enters the analyzer through the sample inlet (1) and then passes through the followings:

- Inlet dust filter (2).
- Measurement chamber located inside the enclosure (9).
- Optional oxygen measurement module (8) if present.

The sample gas pressure is measured at two points:

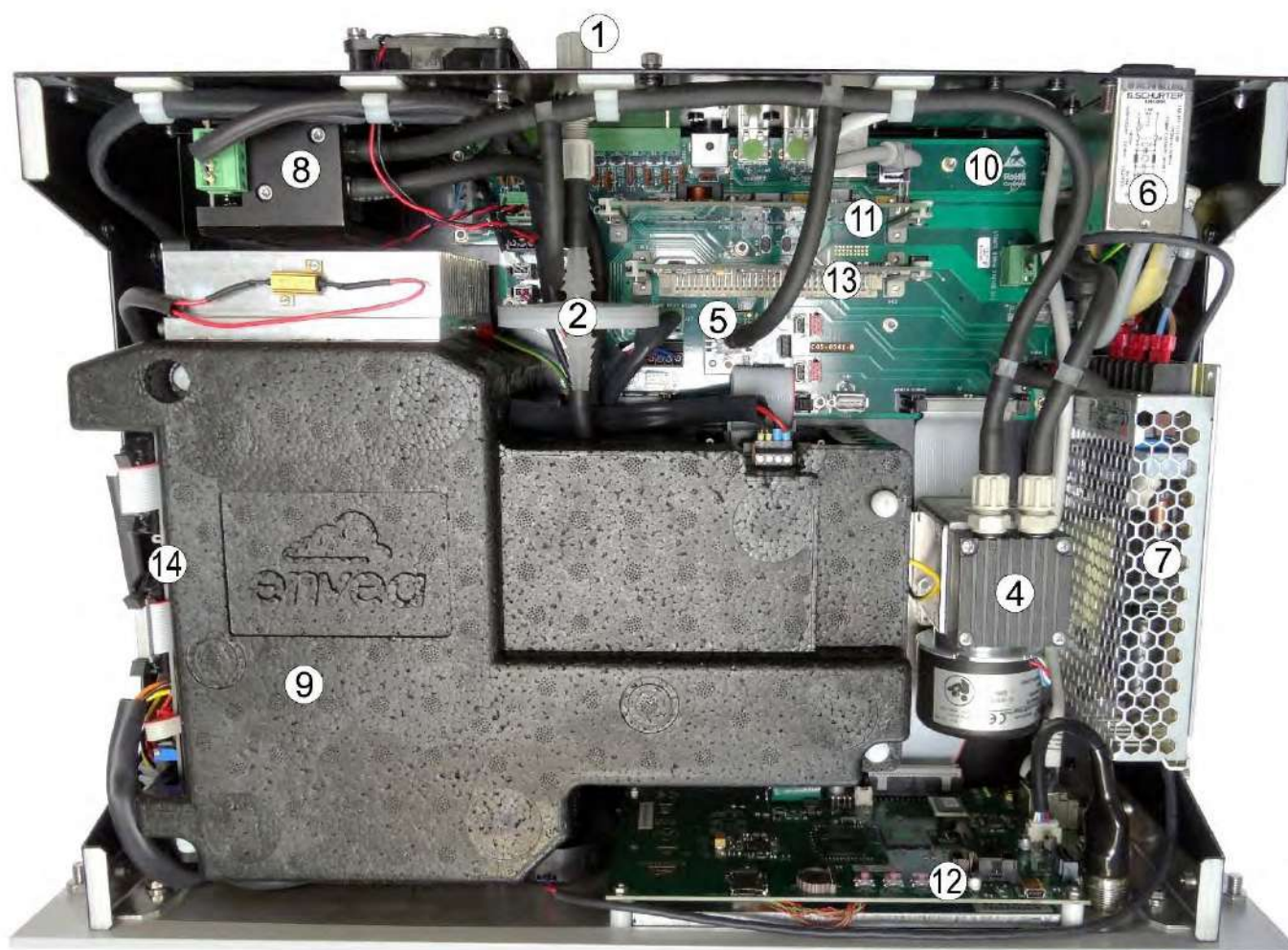
- At the measurement module outlet, by the chamber pressure sensor located under the filter (2), then
- At the pump inlet, by the pressure sensor (5).

The tube located between these two pressure measurement points is fitted with a flow rate restrictor. The pressure difference between the two sensors is thus proportional to the square of the sample flow rate, which can thus be calculated.

The analyzer power supply is protected by the mains filter unit (6), which also has a fuse and an ON/OFF switch.

The mains voltage supplies a 24V power supply unit (7), which then supplies all the electric current required for the analyzer operation.

The Power supply board (11) provides secondary supply voltages (+12V / +5V / +3,3V). The Measurement board (14) drives the measurement module located inside the enclosure (9). The Controller board (13) controls the temperature regulations and the fluid circuit. The Arm20 board (12) drives the whole analyzer, manages the data and the HMI.



(1) Sample inlet, (2) sample inlet dust filter, (4) pump, (5) pump pressure sensor, (6) mains filter, (7) 24VDC power supply unit, (8) O2 probe, (9) measurement enclosure, (10) Interconnection board, (11) Power supply board, (12) Arm20 board, (13) Controller board, (14) Measurement board.

Figure 1–4 – MIR 9000e internal view



WARNING :

- Mains voltage inside
- Presence of moving parts (correlation wheel, fans)
- Presence of hot parts

POSSIBLE OPERATION MODES

- Using the HMI on the front panel touch screen, a PC, a tablet or a smartphone
- Using a serial communication protocol (for example MODBUS)
- Using logical and analog inputs/outputs via the optional modules
- Display measurement values in ppm, mg/Nm³ or %
- Local data storage of average measurement values, with programmable integration time
- Programmable sequences of zero and calibration checks
- Programmable sequences of reference-zero and automatic self-calibration
- Self-monitoring of internal parameters to ensure correct operation

AVAILABLE OPTIONS

The following options are available:

- An O₂ measurement module, by ZrO₂ or Paramagnetic cell
- Up to 6 external solenoid valves, in order to control zero and span checks, zero-references and automatic self-calibrations
- An external sample pump
- A filtering cartridge designed to use the ambient air as zero gas
- One or two input/output modules, to be directly plugged into the USB ports of the rear panel.
- An external NO_x converter oven
- The WEX® software for data acquisition and system handling (DAHS)

CHARACTERISTICS

TECHNICAL CHARACTERISTICS

Measured gases (1 to 7 depending on configuration)	: SO ₂ , NO or NO _x , CO, CO ₂ , O ₂ , N ₂ O, residual H ₂ O in the dry gas
Measurement units (programmable)	: ppm, mg/Nm ³ or % vol.
<u>Measurement ranges:</u>	
• SO ₂	: 0-75 / 0-1500 mg/Nm ³ 0-25 / 0-500 ppm
• NO _x as NO ₂	: 0-100 / 0-5000 mg/Nm ³
• NO	: 0-50 / 0-750 ppm
• CO	: 0-75 / 0-3000 mg/Nm ³ 0-60 / 0-2500 ppm
• CO ₂	: 0-20 / 0-30 % vol.
• N ₂ O	: 0-100 / 0-200 mg/Nm ³ 0-50 / 0-100 ppm
• CH ₄	: 0-100 / 0-200 mg/Nm ³ 0-150 / 0-300 ppm
Residual H ₂ O	: 0-2 % vol.
Accuracy	: ≤ ± 2 % of full scale
Repeatability	: < ± 2 % of full scale
Zero drift	: < ± 2 % of full scale over 30 days
Span gas drift	: < ± 2 % of full scale over 30 days
Linearity	: < ±1 % of full scale
Resolution display	: 10 ⁻³ unit or 0.1ppm
Detection limit	: 0.1- 0.5 ppm
Response time	: ≤ 120 s depends on distance and application
Electric power supply	: 100-250VAC, 50/60Hz + ground
Consumption	: Warm-up: 110W / 160VA Measurement: 50W / 75 VA
Nominal sample flow rate	: ≈ 0.5 l / min.
Sample temperature	: 0°C to +50°C (at analyzer inlet)
Ambient operating temperature	: +5°C to + 40°C
Maximum sample dew point	: Below ambient temperature
Display and HMI	: Color touch screen TFT LCD 7 inches, resolution: 800 x 480 (RGB)
Dimensions (D x L x H) / Weight	: 19" rack, 3U; 483 x 330 x 133 mm 10 kg / 22 lbs

Alarm controls	: Detection and indication of operating malfunctions: temperatures, pressures, currents, voltages, residual moisture level and so on.
Maintenance check-up and diagnostics	: Via touch screen HMI or remotely using embedded web server
Data storage holding time	: 1 year of 1-minute average values
Standard inputs / outputs	: MODBUS TCP, MODBUS RTU (with dongle) UDP protocol, RJ45
Optional inputs / outputs	: 4 or 8 analog outputs (0-1 V, 0-10 V, 0-20 mA, 0- 20 mA, 4-20 mA), not isolated, with common ground and user-programmable scales 6 or 12 static relays, voltage-free, 50V max, 1A max at 24 VDC 4 or 8 opto isolated logical inputs
3 USB ports	: "A" type, 2.0 USB ports (3.0 compatible) 2 ports on the rear panel for connecting options, 1 port on the front panel for the user
Control of external solenoid valves	: 24VDC ; 0,3 A max per solenoid valve Screw terminal connector for up to 5 optional SV: Zero, Span, Span 1, Span 2, Probe
Aux control	: 24VDC ; 0,3 A max Screw terminal connector for an optional external sample pump or for an additional solenoid valve
Control of the external NOx-converter oven	: Screw terminal connector for an optional NOx- converter oven. Temperature reading and PID regulation are then performed by the analyzer
Oxygen O2	: Zirconia (internal); Paramagnetic option (external)
Measuring range	: 0 – 25% by vol O2
Accuracy	: $\pm 0.5\%$ F.S
Resolution display / Detection limit	: 0.01% by vol O2
Analog input for an external O2 probe	: Screw terminal connector for the analog signal of an optional external O2-probe. 0-10V or 4-20mA
Approvals & Certifications QAL 1 certified by the TÜV	:  Tested AMS Regular Surveillance www.tuv.com ID: 000074621

STORAGE CHARACTERISTICS

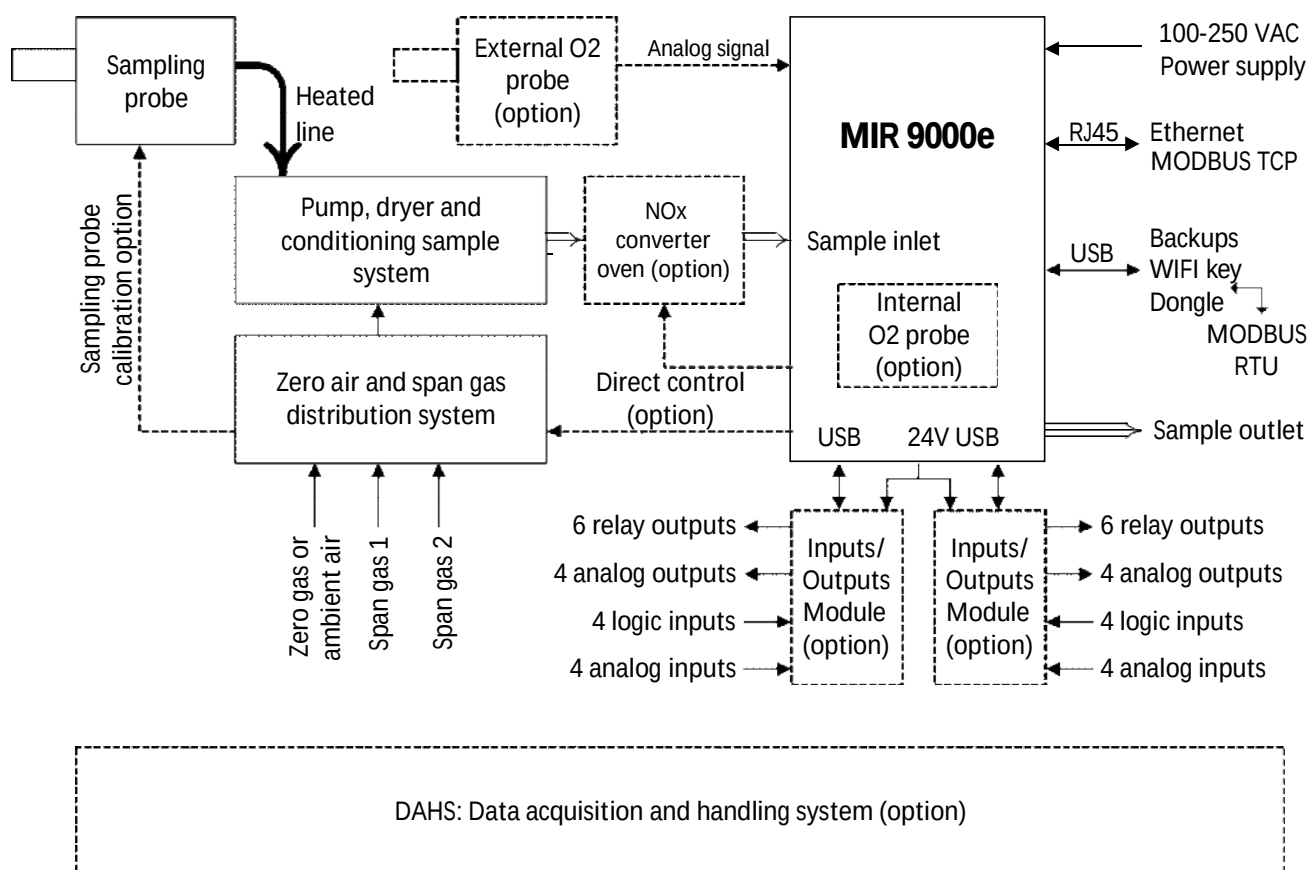
- Temperature: – 10 °C to 60 °C.

INSTALLATION CHARACTERISTICS

Links between units

The MIR 9000e power supplies and external links are shown below:

- MIR 9000e system diagram with condensing dryer (cooling unit type)



- MIR 9000e system diagram with permeation dryer (SEC Probe type)

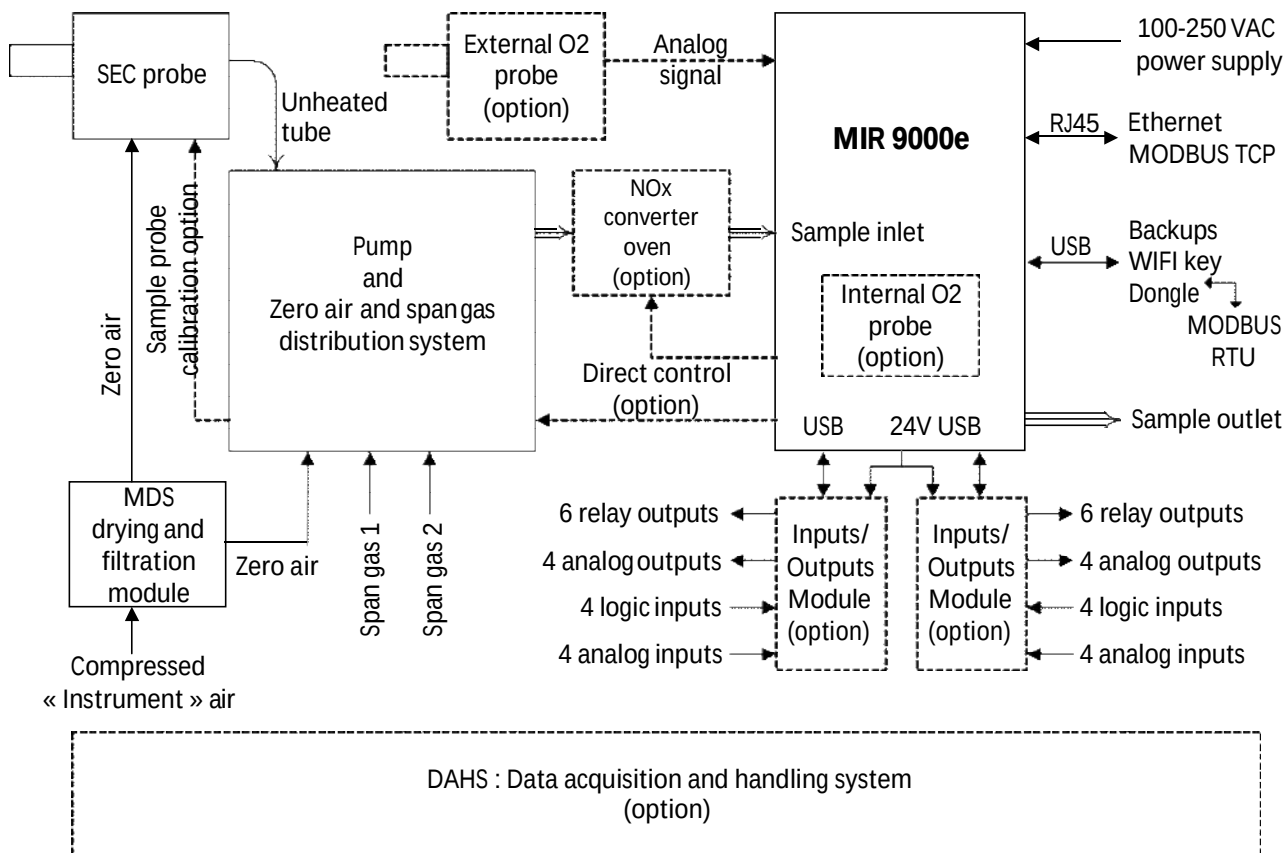


Figure 1-5 – Links between units

Dimensions and weight

The unit comes in a standard 19-inch, three-unit high rack.

Length	: 330 mm (approximately 13 inches) excluding connectors and fittings on the rear panel.
Width	: 483 mm (approximately 19 inches)
Height	: 133 mm (approximately 5.25 inches)
Masse	: 10 kg without option

Handling and storage

The MIR 9000e analyzer must be handled with care to avoid damage to the various connectors and fittings on the rear panel.

The unit must be stored in its original packaging.

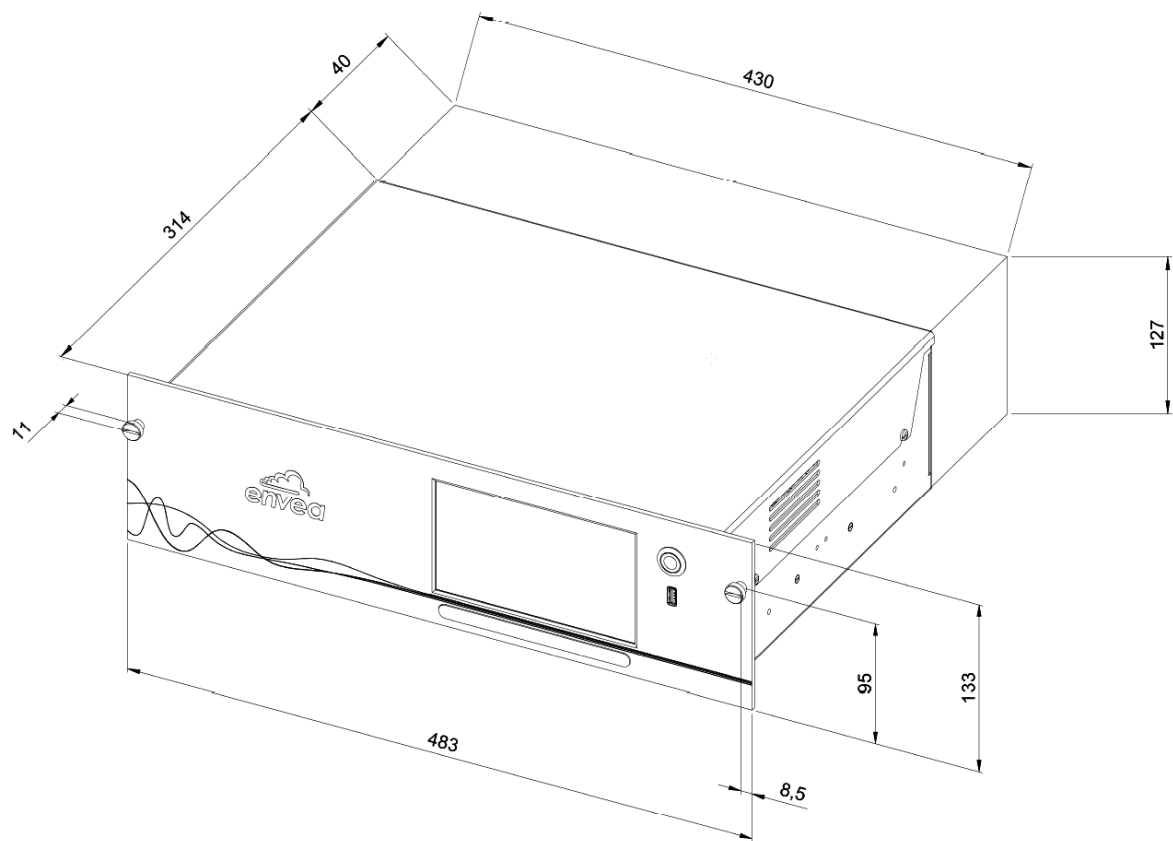


Figure 1-6 – MIR 9000e dimensions (in mm)

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2. OPERATION PRINCIPLE

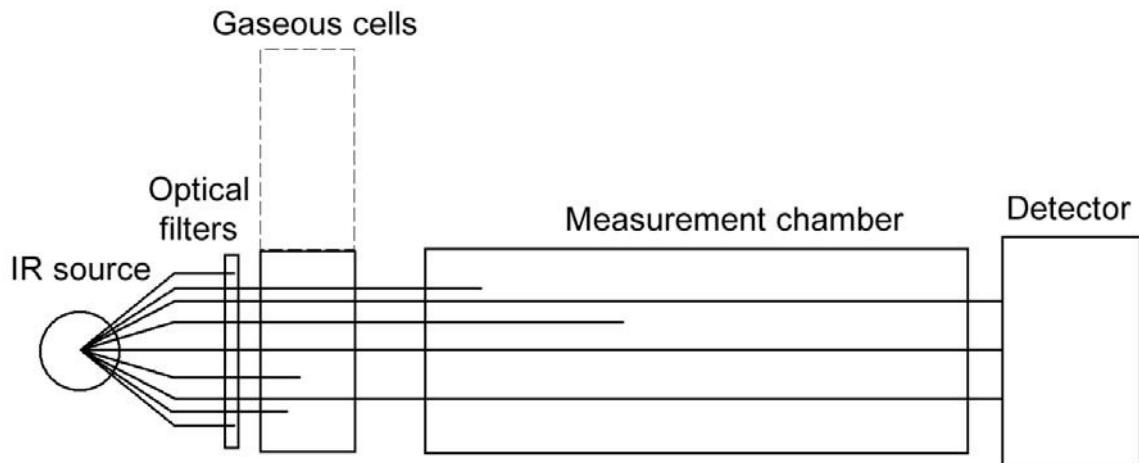


Figure 2-1 – Infrared absorption in the MIR 9000e

INFRARED GAS FILTER CORRELATION SPECTROSCOPY PRINCIPLE

All polyatomic gases absorb electromagnetic radiation at specific wavelengths. Physical analysis based on this phenomenon is called absorption spectroscopy.

The MIR 9000e is a measurement device based on infrared spectroscopy and gas filter correlation. The measurement is based on a purely physical principle, as with almost all our current analyzers.

INFRARED ABSORPTION (FIGURE 2-1)

A beam emitted by an IR source passes through the measurement chamber and is focused on an IR detector. Each gas present on the beam path absorbs the latter at specific wavelengths.

An interference filter which defines a specific wavelength range is placed on the optical path before the measurement chamber.

Let I_0 be the IR intensity received by the detector, when there is no absorbing gas in the measurement chamber, and I the IR intensity received by the detector when absorbing gas is present. The Beer Lambert law results in the following equation:

$$\frac{I}{I_0} = e^{-kLC} \Rightarrow C = -\frac{1}{kL} \ln\left(\frac{I}{I_0}\right)$$

where:

- k is a coefficient depending on the absorbing gas and the wavelength of interest,
- L is the length of the optical path,
- C is the absorbing gas concentration.

Since interference filters define a wavelengths range, rather than a single wavelength, the equation becomes:

$$C = f\left(\frac{I}{I_0}\right)$$

where f is a function of the absorbing gas and the interference filter properties.

This function, f , always resembles a logarithm. For our purposes, it can be approximated by either a polynomial or the logarithm of a polynomial.

It is very difficult to measure I and I_0 simultaneously, as I_0 is the IR intensity received by the detector when there is no absorbing gas. The gas filter correlation technique circumvents this problem.

GAS FILTER CORRELATION PRINCIPLE

A cell filled with a high concentration of the gas to be measured, called the reference cell, is placed on the optical path. The highly concentrated gas in this cell absorbs all of the gas characteristic wavelengths. If some molecules of this gas are present in the measurement chamber, on the optical path, it has no further effect because all the wavelengths they should have absorbed have already been absorbed. As a result, the IR intensity received by the detector can be used as I_R , the reference IR intensity, for this gas.

The infrared detector signal is measured alternately with and without the reference cell on the optical path, by rotating a filter wheel called the correlation wheel. The resulting beam is then absorbed by the gas present in the measurement chamber according to Beer Lambert's law.

The ratio between I , the IR intensity after absorption by the gas, and I_R , the reference IR intensity, is therefore continuously known. The gas concentration can thus be calculated using the formula:

$$C = f\left(\frac{I}{I_R}\right)$$

OTHER ADVANTAGES OF GAS FILTER CORRELATION

The gas in the reference cell "extinguishes" all the spectral lines that are specific to it.

If there are gases on the optical path which are liable to interfere, i.e. which might absorb light in the same wavelength range, it is very unlikely that their absorption lines exactly match those of the gas to be measured. The gases that are liable to interfere will absorb the reference beam and the measurement beam in the same way. Therefore they will have little effect on the I/I_R ratio.

To confirm the gas filter correlation efficiency, let us consider the example of carbon monoxide (CO).

A CO measurement performed by IR absorption without gas filter correlation gives a water vapor (H₂O) sensitivity ratio of about 1/350, despite using an optimal optical interference filter. Thanks to gas filter correlation, this ratio is reduced to only 1/15000, which demonstrates its efficiency.

However, for some gases, gas filter correlation adds nothing more compared to a simple optical reference filter centered on a wavelength not absorbed by the gas to be measured. This is for example the case for the measurement of high concentrations of carbon dioxide (CO₂).

To further improve the metrological characteristics and in particular to eliminate drifts, the instrument periodically readjusts its zero, or in other words, updates the ratio between the reference IR intensity and the measurement IR intensity measured without any absorbing gas in the optical path.

PRINCIPLE APPLIED TO THE MIR 9000e

The clean and dry sample gas is drawn into the multi-reflection measurement chamber by the internal pump. The chamber optical path is 8 meters.

The beam emitted by the infrared source goes across the multi-reflection chamber. This beam is filtered by a series of interference filters and gas cells mounted on a high speed rotating wheel.

A semiconductor detector measures the intensity of the resulting infrared beam after passing through the multi-reflection chamber.

The signal from the detector is amplified and processed electronically and digitally.

The gas concentration is calculated using the ratio of the signals measured in the presence of sample gas to the signals measured in the presence of zero gas (dry air or nitrogen) in the chamber.

The gas concentration values are displayed in real time in the unit previously chosen by the user.

The use of gas filter correlation strongly reduces the cross sensitivities that exist in any traditional infrared measurement.

GENERAL PRINCIPLE DIAGRAMS

COMPLETE SAMPLING AND ANALYSIS SYSTEM

A complete system includes:

- The gas sampling and its transfer from the sampling point to the analysis cabinet
- The gas drying and conditioning unit (filtering....)
- A NO_x converter oven, optionally
- Solenoid valves to inject zero and span gases for automatic checks and self-calibration
- Zero and span gases

There is a large number of installation possibilities, depending on the specificities of each site.

Seven complete system designs are presented here-after. These are natively supported by the MIR 9000e to fit most situations. They correspond to the three "Operating modes" choices indicated in the "Automatic cycle configuration" page.

NOTE REGARDING THE GAS COOLER:

When using a gas cooler, it is possible to use either a single path or a double path heat exchanger. A single path heat exchanger is less costly, but a double one is recommended.

The first heat exchanger, located before the sampling pump, operates at a pressure which is variable and significantly lower than atmospheric pressure. This pressure depends on stack pressure and sampling line pressure loss, which itself depends on flow rate. Therefore, even if the cooler temperature is known and stable, the real sample gas dew point (i.e. at atmospheric pressure) after the first heat exchanger is higher than the cooler temperature, and may vary significantly. Furthermore, depending on sample moisture, ambient temperature, flow rate and heat exchanger design, the real sample gas dew point at the outlet of the exchanger may not be stable.

Using a second heat exchanger located after the sampling pump compensates that. Because it can operate close to atmospheric pressure, at a lower flow rate, on a gas which is already dry with a dew point close to the cooler temperature, the real sample gas dew point at the outlet of the dryer can be more stable, thus improving measurement quality.

The other advantage of using a double path heat exchanger is to prevent cold spots and condensation in the solenoid valve used to inject zero and span gases to the analysis cabinet (see § 3.5 on Calibration). On the other hand, if using a single path gas cooler, the path comprising the solenoid valve between the outlet of the heated line and the inlet of the gas cooler should be vertical and as short as possible.

Although a double path heat exchanger should be favored by design, it is of particular importance when low measurement ranges are required, in particular for SO₂, or when sample gas moisture is high. On the other hand, when sample gas moisture is low, when the lowest measurement ranges are not required, and in the absence of SO₂, choosing a single path heat exchanger can be considered.

“A” TYPE SYSTEM DESIGN

This design must be favoured when pressurized zero gas can be supplied on site, for example in the form of a nitrogen cylinder, as shown.

The examples below display the sampling probe, the heated line, a gas cooler with one (top drawing) or two (bottom drawing) heat exchangers, the sampling pump fitted with a bypass valve for adjusting the sampling flow rate, a NO_x converter oven, the zero gas (nitrogen) cylinder, three span cylinders, and six solenoid valves to drive zero and span gas injections.

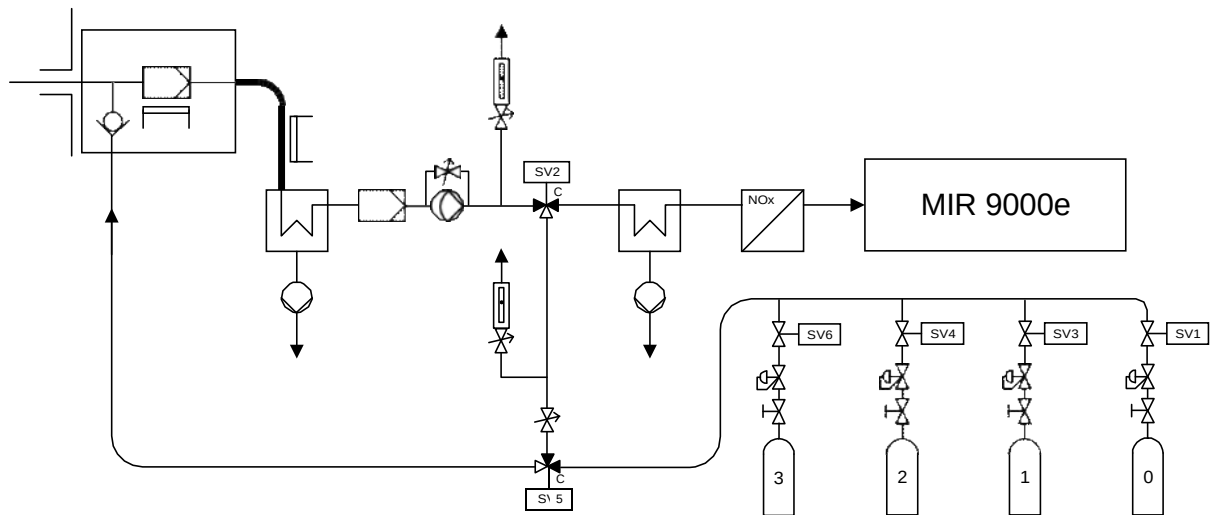


Figure 2-2 – Detailed “A” type system design, with double heat exchanger (best)

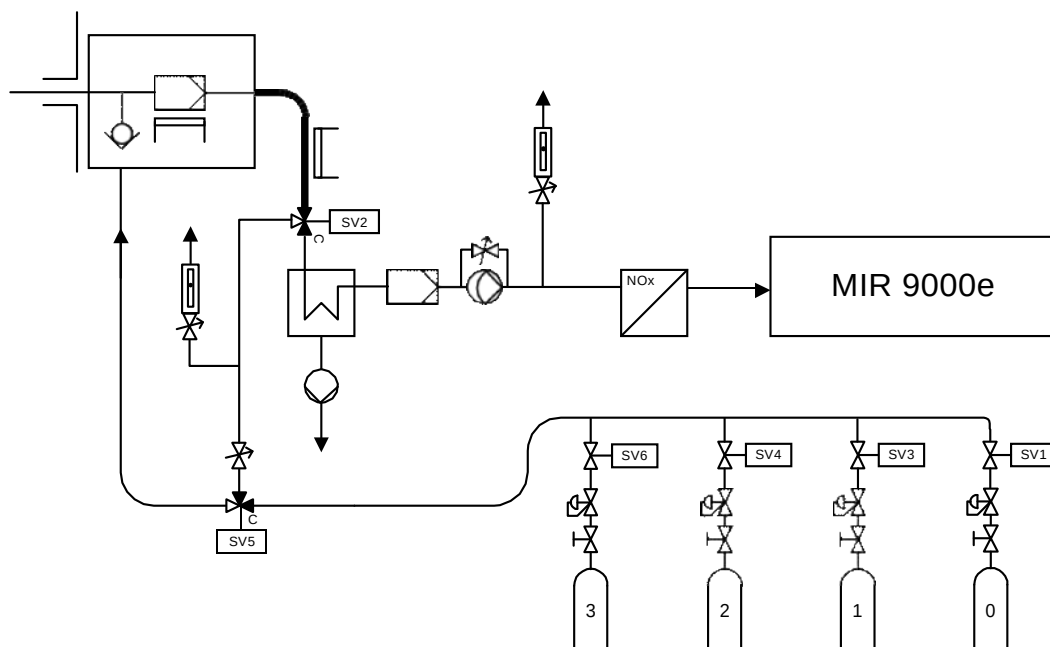


Figure 2-3 – Detailed “A” type system design, with single heat exchanger

“B” TYPE SYSTEM DESIGN

This design must be favoured when pressurized zero gas is not available on site, nor desired.

In that case, the zero gas is supplied by the ambient air via a filtering cartridge and a passage through the cooler to match its dew point with that of the sample gas. Because the zero gas is not pressurized, it can't be sent to the sampling probe, unlike in the “A” type design.

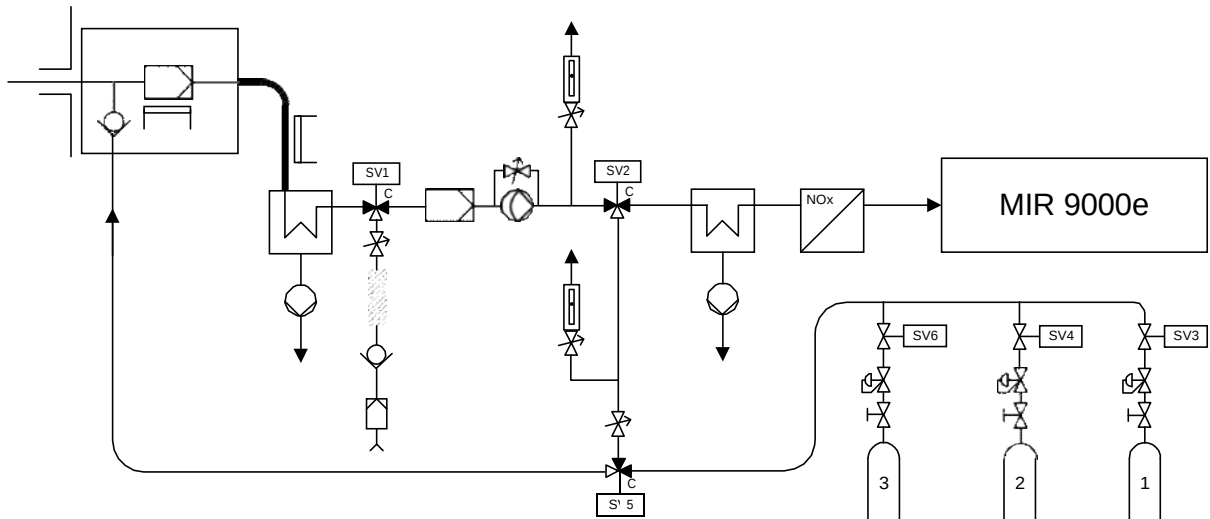


Figure 2-4 – Detailed “B” type system design, with double heat exchanger (best)

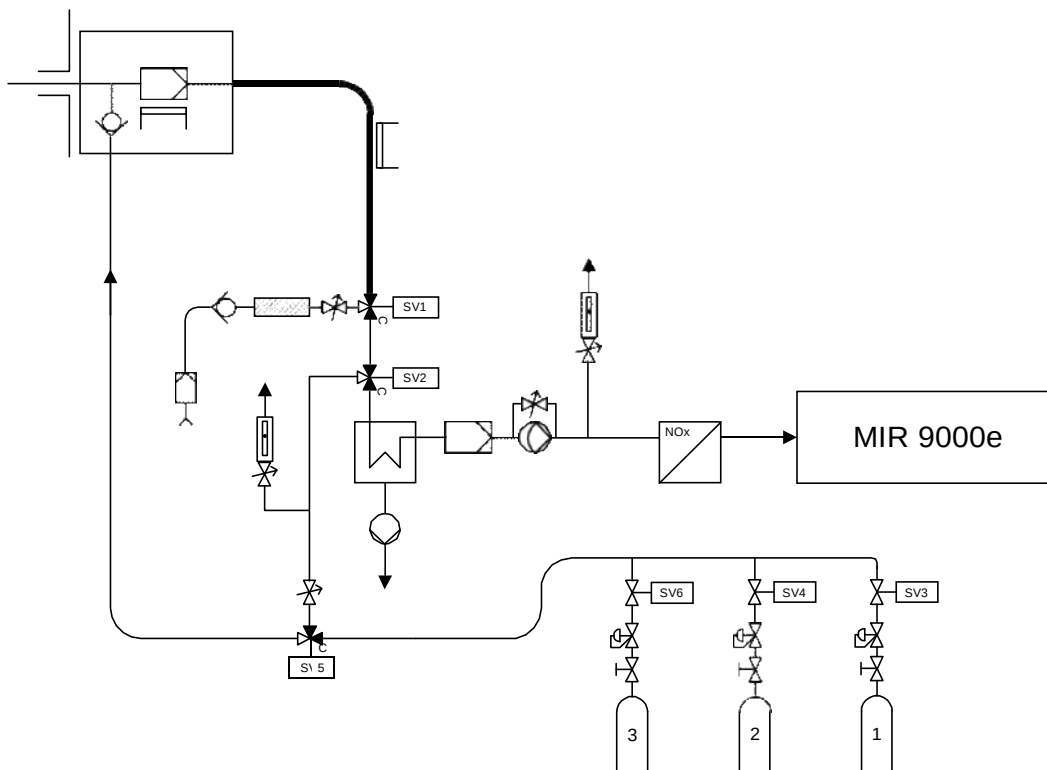
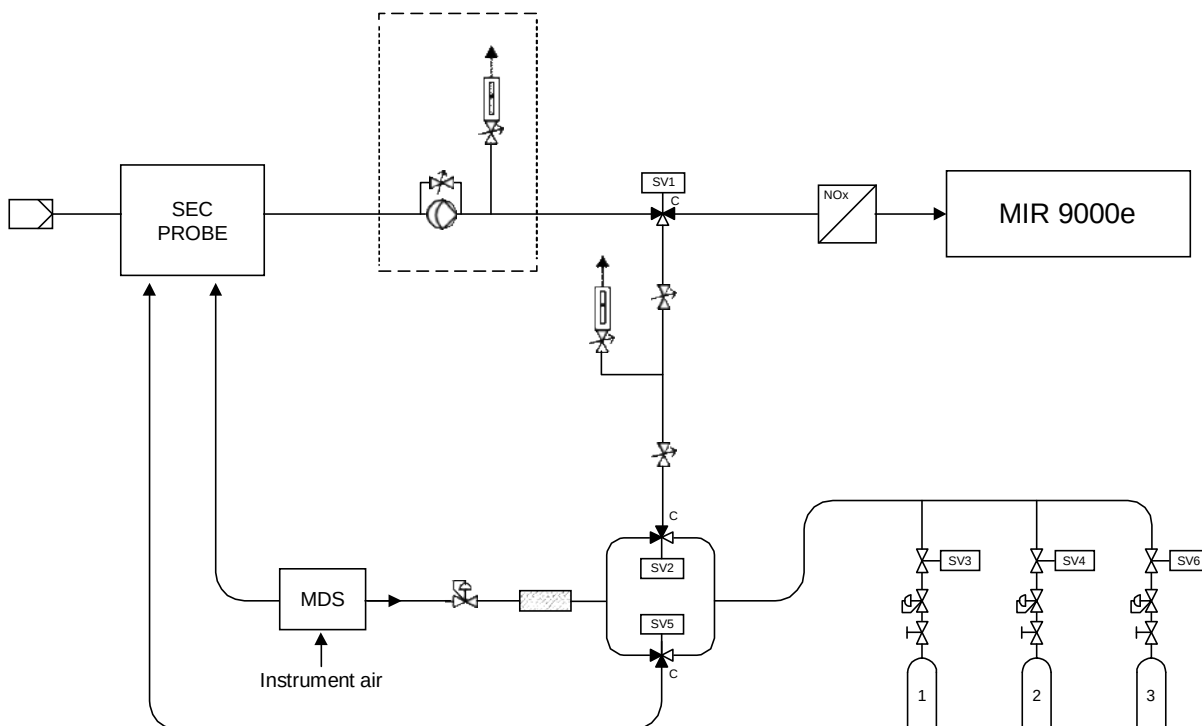


Figure 2-5 – Detailed “B” type system design, with single heat exchanger

“C” TYPE SYSTEM DESIGN

This design is based on the historic architecture of the MIR 9000 LCD / SEC probe / TIG Systems. It allows the use of a permeation dryer (SEC probe type) as an alternative to the condensation dryer (Cooling unit type).

For this reason, dry, oil-free, pressurized instrument air must be available. Three configurations are possible, one with a permeation dryer (SEC probe type), and two with a gas cooler, with either a double or a single heat exchanger.



**Figure 2-6 – Detailed “C” type system design,
with permeation dryer (SEC probe type).**

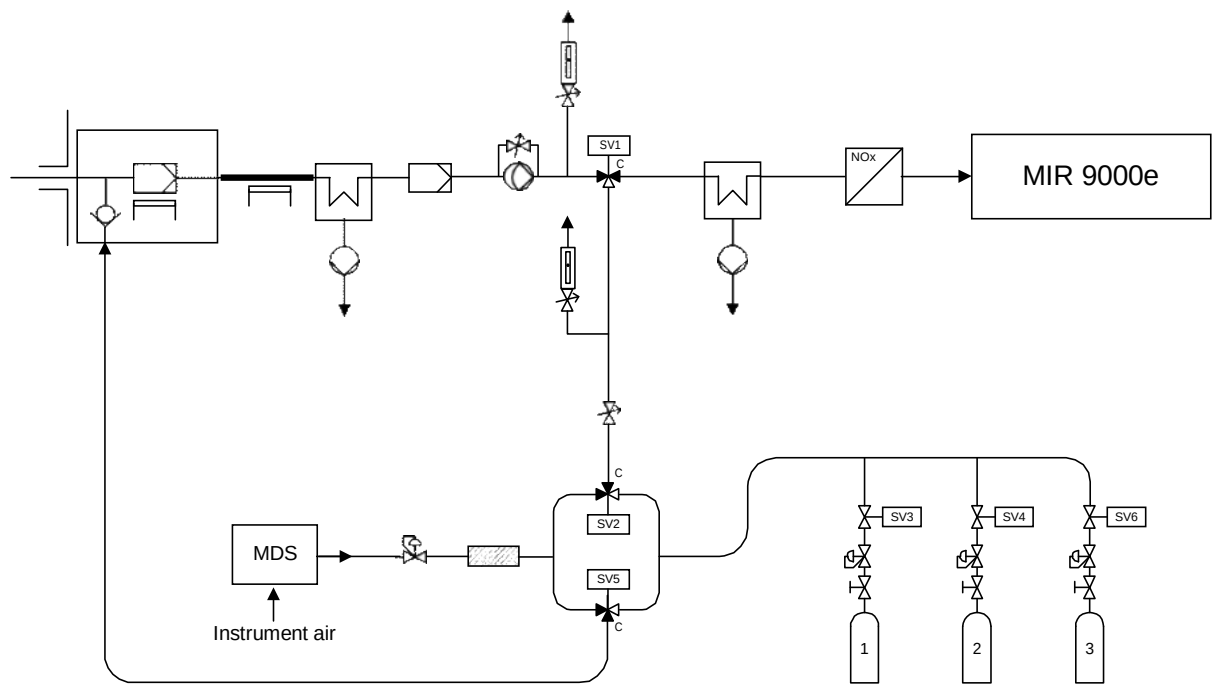


Figure 2-7 – Detailed C type system design, with gas cooler (double heat exchanger)

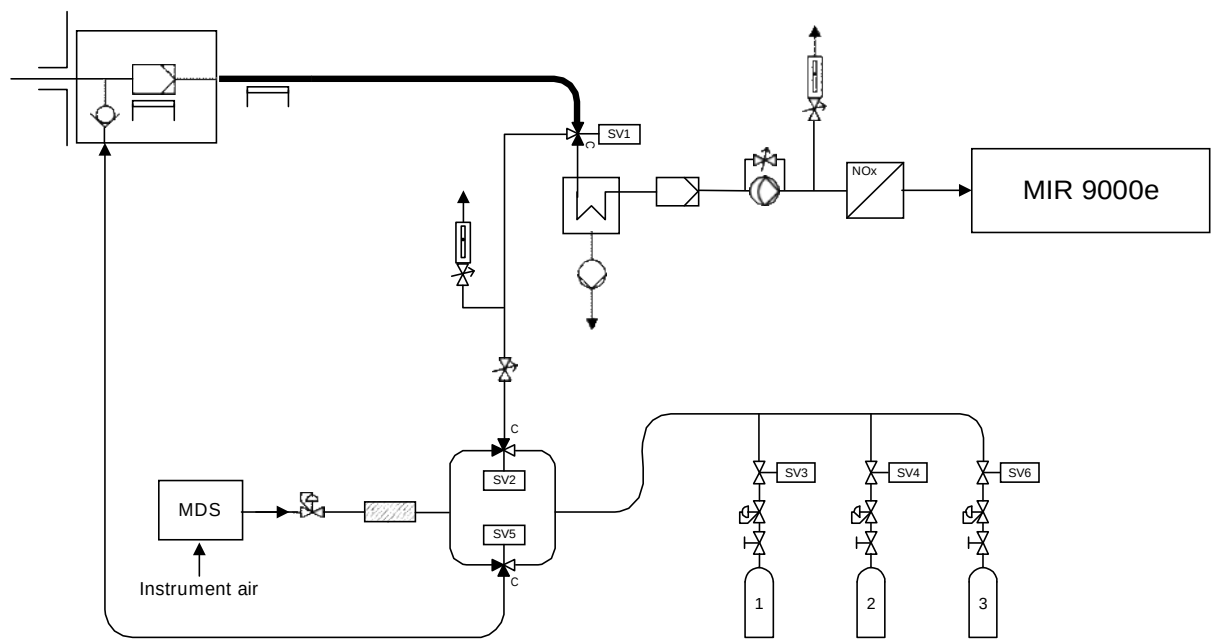


Figure 2-8 – Detailed C type system design, with gas cooler (single heat exchanger)

SIMPLIFIED SYSTEM DIAGRAMS DISPLAYED IN POP-UP HELP WINDOWS OF IHM SOFTWARE

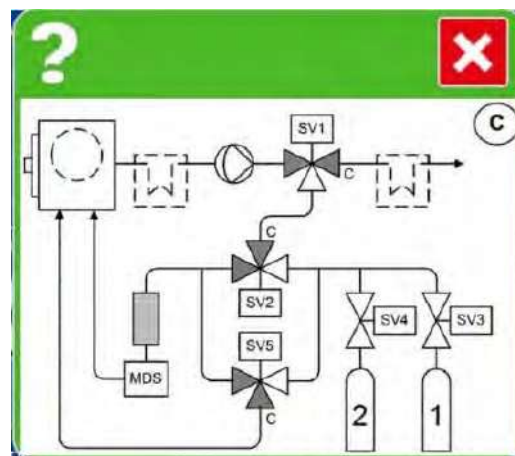
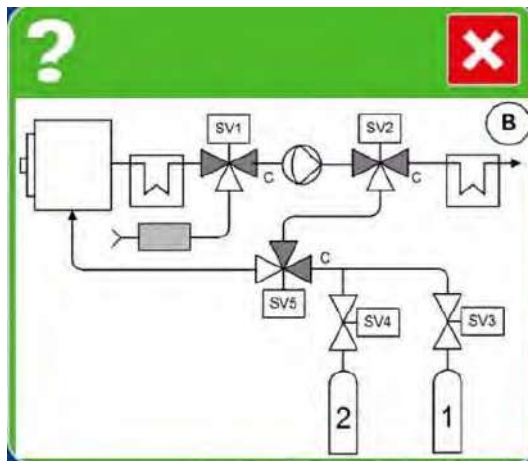
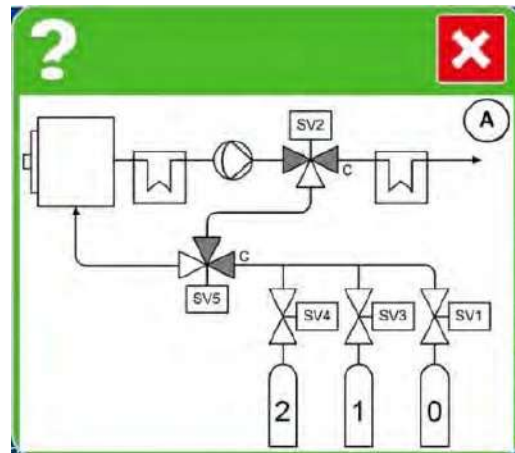


Figure 2-9 – Simplified system diagrams displayed in pop-up help windows of IHM software

MIR 9000e SYNOPTIC DIAGRAM

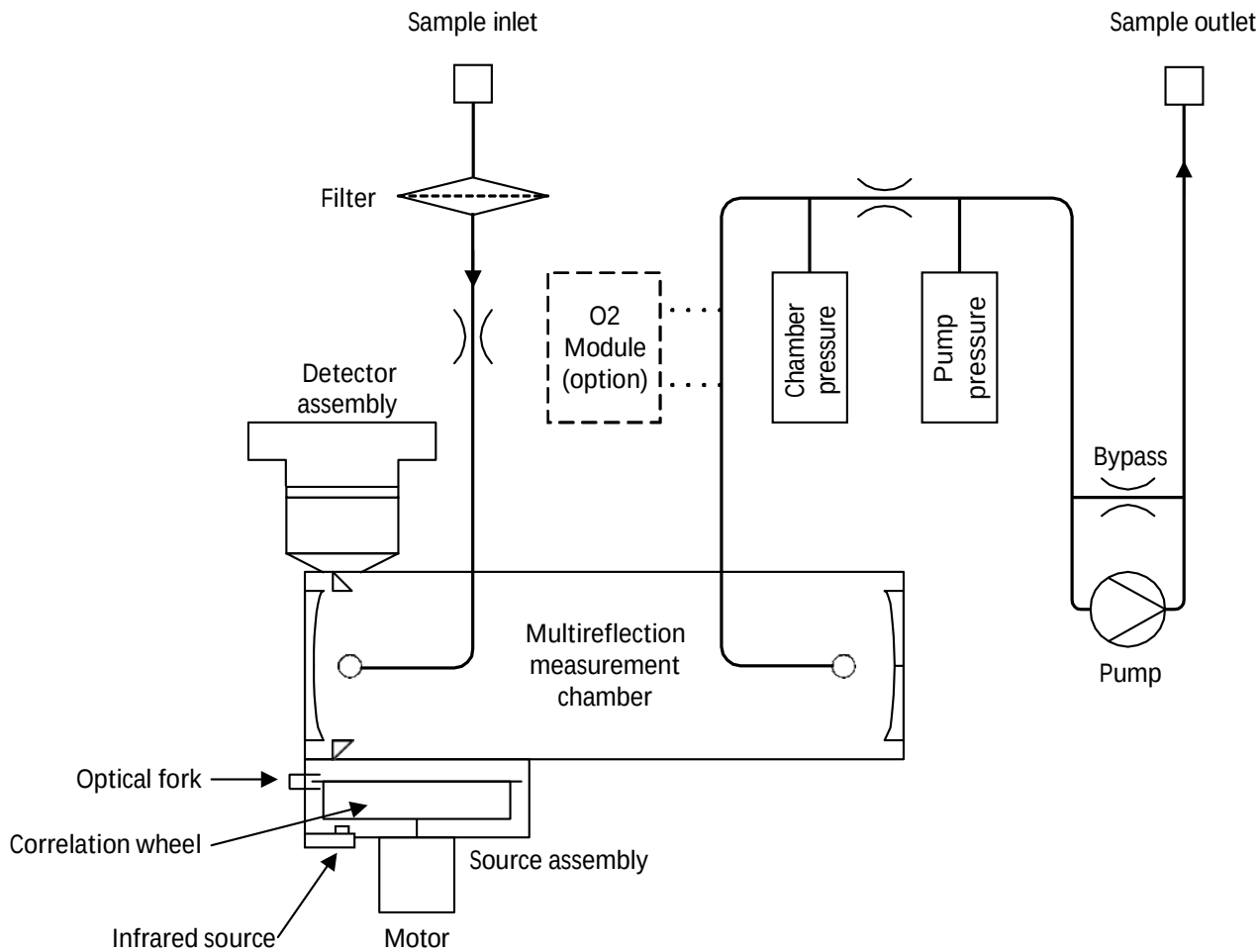


Figure 2-10 – MIR 9000e synoptic diagram

2.2.3

MIR 9000e INTERNAL ARCHITECTURE

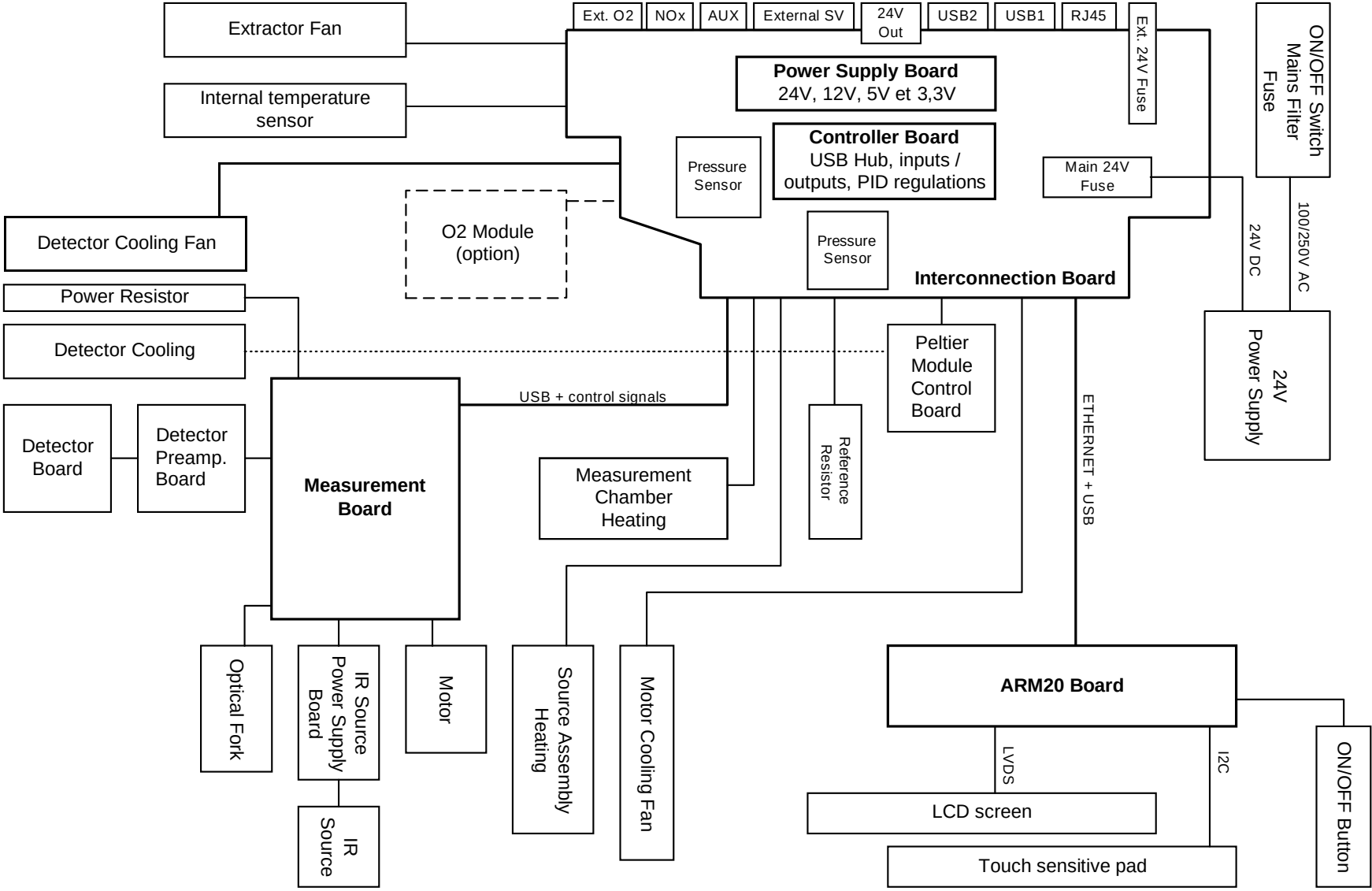


Figure 2-11 – MIR 9000e internal architecture

2.2.4 OPERATION FLOWCHART

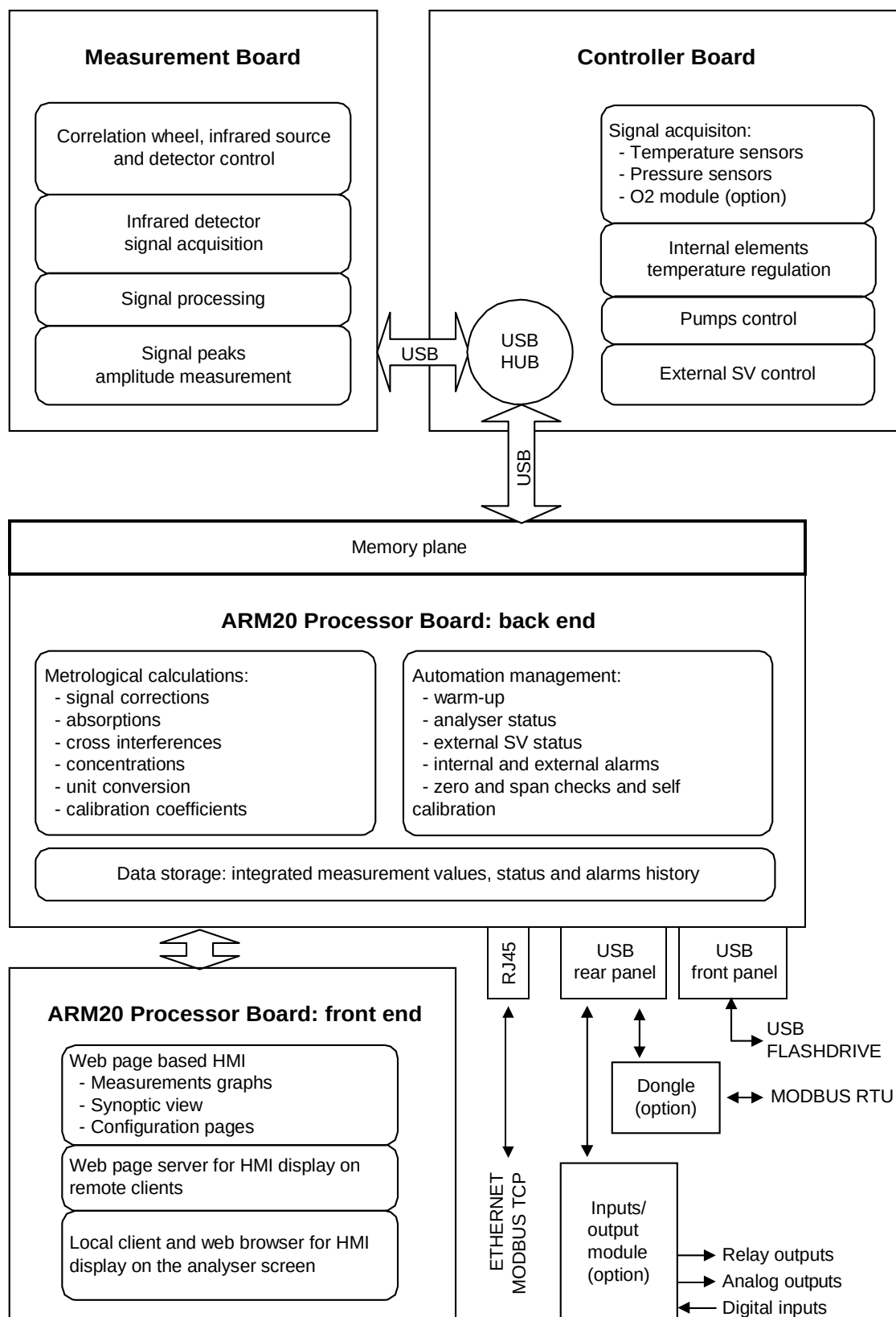


Figure 2-12 – Operation flowchart

MIR 9000E MAIN ELEMENTS DESCRIPTION

The main elements of the MIR 9000e are shown on Figure 2–10 and Figure 2–11.

The analyzer mainly consists of:

- A temperature-controlled optical bench comprising :
 - An infrared source
 - A correlation wheel
 - A multi-reflection measurement chamber
 - An infrared detector
- A sample filter
- A pump
- Two pressure sensors located on either sides of a flow restrictor, which measure at the same time the measurement chamber pressure and the sample flow rate
- An optional oxygen measurement module, based either on a ZrO₂ cell or a paramagnetic cell
- A 24 V power supply unit
- The electronic boards described in section 2.4

OPTICAL BENCH

The optical bench comprises an infrared source, a correlation wheel, a multi-reflection measurement chamber and an infrared detector. It is assembled inside a foam unit which holds the part together, insulates them and protects them against shocks and vibrations.

The infrared source is continuous. It is built around a heated filament. It emits a broad black body type spectrum. The infrared source and the correlation wheel are located in the source assembly which also includes the motor driving the correlation wheel and an optical fork used to encode the wheel position. The source assembly is temperature-controlled at 50°C.

The correlation wheel is fitted with interference filters and gas cells (12 positions). Its rotation speed is regulated at 1500 rpm using a brushless motor.

The multi-reflection measurement chamber is 20 cm long, with an 8 meters long optical path. It is temperature-controlled at 50°C. Its gas inlet is fitted with a deflector whose role is to homogenize the gas flow throughout the volume of the measurement chamber.

The detector is of semiconductor type with high detectivity. It is cooled by three Peltier stages. The first stage regulates the assembly where the detector and its preamplifier board are located at 25°C (as standard).

The detector infrared sensitive chip is cooled at -35°C (as standard) by a double-stage Peltier element located inside the detector casing.

SAMPLE FILTER

The sample filter is an in-line PTFE membrane filter, located between the sample inlet and the measurement chamber. It aims to protect the analyzer against particles and aerosols greater than 0.45µm (as standard). It can be replaced by other filter types, such as water-stop filters, for enhanced protection.



REMARK: The sample gas must be clean when it arrives at the analyzer inlet. In no way can the analyzer sample filter replace the filters normally present in the sampling system.

ANALYZER PUMP

The analyzer pump is located between the measurement chamber outlet and the analyzer sample outlet. It is a diaphragm pump. As the pump is located downstream the measurement chamber, it has no memory effect on the measurement. The materials in contact with the sample gas are listed in the table below:

	Standard pump	On request
Pump head	PPS	PPS
Diaphragm	EPDM	PTFE
Valves	FPM	FFPM

The pump is driven by a 24V PWM power supply. In normal operation, pump speed is controlled in order to regulate the measurement chamber pressure. A set of flow restrictors located on either side of the measurement chamber, and a bypass flow restrictor located between the pump inlet and outlet, ensure a suitable flow of approximately 0.5 l/m while allowing the pump to run at an appropriate rating.

The measurement chamber pressure regulation stabilizes the sample gas spectroscopic properties, independently of stack pressure, atmospheric pressure and clogging of the sampling system filters.

The analyzer pump is primarily used to draw the sample in from the analyzer inlet, and to control the pressure in the measurement chamber. Its role is not to replace the sampling system pump, although this can be done in specific cases. It is always better to separate the roles of each pump:

- The role of the sample pump is to transfer the sample gas from the sampling probe to the analysis cabinet, at a sufficient flow rate to optimize the response time. It must be able to overcome long distances and filter clogging.
- The role of the analyzer pump is to draw the sample into the measurement chamber and to regulate the pressure in the measurement chamber independently of external conditions.

PRESSURE SENSORS

The two pressure sensors are directly located on the Interconnection board. One is used to measure the measurement chamber pressure and the other one is used to measure the pressure at the analyzer pump inlet.

They are located on either sides of a flow restrictor. The pressure difference between the two sensors is proportional to the square of the flow rate of the gas passing through the restrictor. This property is used by the MIR 9000e to measure the sample gas flow rate.

OXYGEN MEASUREMENT MODULE (OPTIONAL)

An optional oxygen measurement module can be integrated into the MIR 9000e. When present, it is located immediately downstream of the measurement chamber.

Two types of oxygen measurement modules are available: ZrO₂ and Paramagnetic. These optional modules can be integrated into the analyzer at any time.

These two modules do not directly measure the oxygen concentration. They measure partial oxygen pressure. The oxygen concentration is then obtained by dividing the partial pressure by the total pressure of the sample gas.

The MIR 9000e can also integrate the oxygen measurement performed by an external probe, using the dedicated analog input located on the rear panel (see figure 1-3).

24 V POWER SUPPLY

The 24V power supply unit powers the whole analyzer and also, if necessary, the elements connected to the rear panel: external solenoid valves, external 24V sample pump, and input/output modules. The 24 V power supply is the only analyzer element connected to the mains power supply.

Its rated input current is 3A at 115 VAC and 1.7A at 230 VAC. Its input is protected by the 3.15A fuse located in the mains power supply filter on the rear panel.

Its rated output current is 6.5A. It is protected by the 8A time-delayed fuse located on the Interconnection board (Ø5 x 20 mm format).

The 24V outputs on the rear panel are also protected by a second fuse (2A, fast type), accessible from the rear panel of the analyzer (Ø5 x 20 mm format).

MIR 9000E ELECTRONIC ARCHITECTURE DESCRIPTION

The communication between the ARM20 board and the other electronic boards (measurement board, interconnection board, power supply board, controller board, pressure sensor boards, etc.) is carried out according to the diagram shown in Figure 2–11 and Figure 2–12.

The ARM20 board, named by reference to its microprocessor, is a fast computation and interfacing (communication) board designed for the "e" series analyzers. It is mounted on all analyzers and offers an Ethernet output (RJ45 socket), and three USB ports (one on the front panel and two on the rear panel via the Interconnections board). For these analyzers, the ARM20 board is the central element for electronic and metrological operation and external communication.

The MEASUREMENT board drives the infrared measurement module. It has two microcontrollers. It acquires the signals and operating parameters of the measurement module. These data are pre-processed and transferred to the ARM20 board via a USB communication protocol.

The CONTROLLER board has one microcontroller. This board ensures the following main functions:

- Temperature sensor acquisition and power regulation of the internal heating elements
- Pressure sensor signal acquisition
- Optional O2 module signal acquisition, if applicable
- External solenoid valves power supply and switching (+24 V)
- Pump control
- USB hub used for communication between the different boards

The POWER SUPPLY board generates several internal analyzer voltages from the 24V power supply unit:

- +24 V filtered and secured
- +12 V
- +5 V
- +3.3 V

The INTERCONNECTION board interfaces the Power supply, Arm20, Controller and Measurement boards, and connects various electrical devices and sensors to each other.

In addition to the standard e-Series electronics, the MIR 9000e comprises the following specific boards:

- The DETECTOR board, located in the detector assembly, which simply holds the infrared detector.
- The DETECTOR PREAMPLIFIER board, also located in the detector assembly, which pre-amplifies the detector signal. The DETECTOR and DETECTOR PREAMPLIFIER boards are assembled one over the other.
- The PELTIER MODULE CONTROL BOARD which converts the PWM signal from the Controller board into the direct current which powers the Peltier module of the detector assembly cooling element. It also sends the detector assembly temperature signal back to the Controller board.
- The INFRARED SOURCE POWER SUPPLY board, located on top of the measurement chamber, which powers the infrared source. This power supply operates through current regulation. It also holds a power transistor used to power the internal Peltier element of the infrared sensor.

CHAPTER 3

OPERATING INSTRUCTIONS

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Figure 3–1 – MIR 9000e rear panel

3–4

3. OPERATING INSTRUCTIONS

INITIAL STARTUP

The analyzer is checked and calibrated before delivery. Calibration is checked in factory.

PRELIMINARY OPERATIONS

Refer to Figure 3–1.

It is necessary to leave a free space of at least 10 cm (4 in.) between the rear panel and the wall behind the analyzer to ensure correct analyzer ventilation.

Starting up the analyzer involves the following preliminary operations:

- Visually inspect the inside of the instrument to make sure no damage occurred during transport.
- Connect the analyzer sample inlet (1) with a 4/6 mm Teflon tube.
- Connect the analyzer sample outlet to vent (2) with a 4/6 mm Teflon tube.
- If necessary, according to the available options, connect :
 - The ETHERNET cord (3)
 - The input/output module(s) via the USB ports on the rear panel (4) and the dedicated 24V power supply socket (5)
 - The 24V solenoid valves (6):
 - The « Zero » solenoid valve to SV1
 - The « Span » solenoid valve to SV2
 - The « Span 1 » solenoid valve to SV3
 - The « Span 2 » solenoid valve to SV4
 - The « Probe » solenoid valve to SV5
 - The 24V external pump or the « Span 3 » solenoid valve to AUX (7)
 - The optional dedicated NOx-Converter oven (8)
 - An external O2-Probe (9)
- Using the mains power cord, connect the analyzer power plug (10) to a mains socket fitted with ground.

NOTE : The analyzer is equipped with a power supply compatible with a 100-240V voltage/50-60 Hz frequency mains.



(1) sample inlet, (2) sample outlet, (3) Ethernet port, (4) USB port, (5) 24V power supply connection of the Input / Output Module option, (6) 10-point BL connection to connect 1 to 5 external calibration solenoid valve(s), (7) 2-point BL auxiliary connection to connect a 24V external pump or an additional external solenoid valve, (8) 4-point BL connection to connect a dedicated NOx converter oven, (9) 3-point BL connection to connect the analog signal of an external O2-probe, either voltage or current type (not both at the same time, because this is a single input), (10) three-contact socket for mains connection via a standard cord, (11) ON/OFF switch, (12) 3,15 A / 250 V general fuse, 5x20 mm size, time delay type, (13) 24V-2A fuse to protect the (5), (6) and (7) outputs, (14) extractor fan.

Figure 3-1 – MIR 9000e rear panel

STARTING UP THE ANALYZER

To start-up the analyzer, carry out the INSTALLATION/COMMISSIONING procedure described below in the following order.

1/ First press the ON/OFF switch on the rear panel, then the push button on the front panel. The analyzer starts running, and begins the warm-up cycle (duration of this cycle is 1800 seconds maximum). The home page is displayed: it shows the warm-up progress.



Warm-up finishes when all internal parameters are within the operating limits. The warm-up icon then disappears.

If the « Zero Ref. at startup » field was set to ON in factory in the « Automatic cycle configuration » page, the analyzer performs a zero-reference automatically:



When the zero reference ends, the zero reference icon disappears and the following page is displayed.



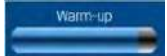
This page is the **standard functions** home page of the analyzer. It contains additional pages displayed by touching / clicking on the  top buttons. These pages display the measurement channels.

Touch / click on  to display the keys of the standard functions. These keys are described in the table below.

From the touch screen mounted on the front panel: no password is needed to access the home page.

From a remote computer: a password is required in the  field. Use one of the passwords indicated hereafter in § 2/ and validate with .

Description of the status and menu bar present at the screen base

	The left button shows the fluid circuit status and displays the gas injection controls:  Sample gas measurement,  Zero gas injection in the analyzer cabinet,  Zero gas injection in the sampling probe,  to : Sample gas injection (cylinder 1 to 5) in the analyzer cabinet,  to : Sample gas injection (cylinder 1 to 5) in the sampling probe.
	Status icon location. Status icons are the following:  Shows that the analyzer operates normally,  Shows that the analyzer is warming-up,  Shows that the analyzer is in alarm,  Shows that the analyzer is disconnected from the network,  Shows that the analyzer is in Maintenance mode,  Shows that the analyzer is on standby.
	Information area indicates model and current analyzer time.
	Displays warm-up progress.
	Stops the cycle in progress.
	Indicates password level and displays a pop-up window to modify this password (as shown below). 
	Displays the pop-up help.
	Displays the analyzer information panel.

	Displays the automatic cycles manual controls:  Heads zero and span gases to the sample probe,  - -  Launches a check-up cycle on zero air, span 1, span 2 ... up to span 5  - -  Launches an automatic self calibration cycle on zero-air (zero-reference), span 1, span 2... up to span 5  Heads the zero and span gases to the analyzer cabinet.
	Displays the standard features buttons :  Accesses the "Find and display recorded averages" function  Accesses the "History" function  Accesses the "Diagnostic functions"  Accesses the "Configuration menu".  Displays the analyzer synoptic diagram  Displays real-time graphs of selected measurement channels  Displays the measurement channels as lines,  Displays the measurement channels as bar-graphs.
	Accesses the analyzer home page.

2/ Access to the analyzer configuration menu

From the touch screen mounted on the analyzer front panel:

Touch  to display the password input window , then touch the field  to display the touch keyboard  in the lower half of the screen.

Touch  to switch from alphanumeric to numeric keyboard if necessary. Enter the password and validate with : the password window and keyboard close, and the configuration menu opens.



Touching  closes the password window and cancels the request.

From a remote computer: do the same with the computer keyboard and mouse.

Access to the configuration functions depends on the hierarchical level of the user's password. There are three hierarchical levels: User, Advanced, and Expert. **The factory passwords available in the analyzer on delivery are the followings, they are symbolized by the icons:**

Password level	Login	Icon
User	12345	
Advanced	78300	
Expert	00007	

At initial start-up, it is strongly advised to change the factory passwords: use  in the configuration functions menu to display the password management window, input the new passwords in the corresponding fields, then use  to validate the input and close the password management window.

The configuration functions access menu is shown below.

The allowed functions are white on blue background (refer to area 2). As explain above, they depend on password level. They are contact-activated. When a function is activated, it changes color from blue  (e.g.) to green  (e.g.), while its name is displayed in area 1 at the top and  is activated to . By clicking again on  (e.g.), or activating , the user accesses the corresponding function page.



3/ Pop-up help operation

The pop-up help is obtained by touching / clicking on  which turns green . Then, touch / click on buttons (e.g. ) or display areas to read about them: a pop-up  opens to give the desired information. The pop-up help closes by touching / clicking on  .

NOTE : The pop-up help is disabled by touching / clicking on  again.

See the example below:



1 button selected to know the operation or use, 2 message giving the desired information.

3 button used to close the pop-up help.

PROGRAMMING THE MIR 9000E

SCREEN AND KEYBOARD DESCRIPTION AND USER INSTRUCTIONS

The screen mounted on the front panel is touch sensitive. Most features can be used simply by touching control buttons displayed on the screen. When the user is required to enter a parameter, a virtual keyboard pops up.

When the analyzer is connected to a remote computer, parameters are modified using the computer keyboard, and control buttons are used by clicking on them with the mouse.

Control screen



The control screen areas are defined as follows:

1	Browsing bar for multi page menus. Use the buttons to select the desired page.
2	Measurement or configuration area. This area displays the measurement parameters (gas, value, units...) or the configurable parameters associated with the selected page.
3	Manual controls, information and browsing bar. This bar is used to control gas injection, zero reference and self calibration cycles. It displays the analyzer status, the information panel icon and the standard features menu icon.

The manual controls, information and browsing bar 3 is available on all pages. Details on browsing ergonomony are given in section 3.3.

Keyboard input

A virtual touch-sensitive keyboard is displayed instantly in the lower half of the screen [2] when the user touches a field to be filled in or modified.

Two keyboard types are available: numeric keyboard if numeric entry is required or QWERTY keyboard if alphanumeric entry is required.

Examples below show the keyboard display in the « *General configuration* » page.

When the user touches any keyboard key, button [1] becomes . It is used to validate entries and close the virtual keyboard. Button [3] is used to close the keyboard without validation.

NOTE : Field inputs are indexed.

These keyboard keys work like Android system keys found on tablets and smartphones.

The numeric keyboard is shown below [2]:

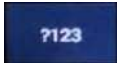
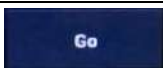


The alphanumeric keyboard is shown below (2):



(1) entry validation button, (2) numeric keyboard with touchscreens, (3) button to close keyboard without validation.

Virtual keyboard specific keys:

	Switches from alphanumeric to numeric keyboard.
	Switches from numeric to alphanumeric keyboard.
	Switches from numeric to symbol keyboard.
	Enters inputs.
	Clears inputs.

From a remote computer, use the keyboard and mouse.

OPERATING PARAMETERS PROGRAMMING

Two possible fields are available to enter or change parameters.

Fields requiring character input

In front of the analyzer touchscreen, touch the input field to be modified. The virtual input screen is displayed with the QWERTY keyboard if alphanumeric entry is required, or the numeric keyboard if numeric entry is required. If letters and numbers are required in the same field, touch the **ABC** key to switch from numeric to alphanumeric keyboard, and the **?123** key to switch from alphanumeric to numeric keyboard. Input with the virtual touch keys, enter by touching the **Go** button, and validate with . The virtual keyboard closes and the modified field is displayed properly filled in.

When modification is done from a remote computer, the computer keyboard and mouse is used.

Fields requiring parameter selection in a list

Touch the field to be modified. The list of parameters or values to be selected is displayed with the current parameter or value in blue background. Touch the desired parameter or value to select it. The new parameter or value is displayed in blue background. Validate with . The list closes, and this new parameter or value replaces the previous.



ON/OFF buttons

To switch OFF to ON, touch (or click on) the OFF button. The ON field becomes white in blue background. And vice versa.



NOTE : ALWAYS validate any parameter change with  so that the analyzer takes it into account.

BROWSING ERGONOMY

The MIR 9000e home page is the following:



The home page displays up to five measurement channels.

From the touch screen mounted on the analyzer front panel, the home page gives access to the standard functions without using a password.

From a remote computer, the following page is displayed first:



The user enters the password in field (1) and validates with  to open the home page.

The home page is composed of two browsing bars:

- Top browsing bar

Change page by using the  buttons or sliding the page to the left or right.



- Base browsing bar

As indicated in section 3.2.1.1, this bar is used to control gas injection, zero reference and self calibration cycles. It displays the analyzer status, the information panel icon and the standard features menu icon. This bar is available in all pages.



Description of the status and menu bar present at the screen base (the same than on pages 3-8 and 3-9)

	The left button shows the fluid circuit status and displays the gas injection controls:  Sample gas measurement,  Zero gas injection in the analyzer cabinet,  Zero gas injection in the sampling probe,  to : Sample gas injection (cylinder 1 to 5) in the analyzer cabinet,  to : Sample gas injection (cylinder 1 to 5) in the sampling probe.
	Status icon location. Status icons are the following:  Shows that the analyzer operates normally,  Shows that the analyzer is warming-up,  Shows that the analyzer is in alarm,  Shows that the analyzer is disconnected from the network,  Shows that the analyzer is in Maintenance mode,  Shows that the analyzer is on standby.
	Information area indicates model and current analyzer time.

	Displays warm-up progress.
	Stops the cycle in progress.
	Indicates password level and displays a pop-up window to modify this password (as shown below). 
	Displays the pop-up help.
	Displays the analyzer information panel.
	Displays the automatic cycles manual controls:  Heads zero and span gases to the sample probe,  --  Launches a check-up cycle on zero air, span 1, span 2 ... up to span 5  --  Launches an automatic self calibration cycle on zero-air (zero-reference), span 1, span 2... up to span 5  Heads the zero and span gases to the analyzer cabinet.
	Displays the standard features buttons :  Accesses the "Find and display recorded averages" function  Accesses the "History" function  Accesses the "Diagnostic functions"  Accesses the "Configuration menu".  Displays the analyzer synoptic diagram  Displays real-time graphs of selected measurement channels  Displays the measurement channels as lines,  Displays the measurement channels as bar-graphs.
	Accesses the analyzer home page.

The user displays the configuration function access menu by touching / clicking on the  button. The user-accessible function buttons are white-framed in blue background. They depend on the password level assigned to the user. Each button provides access to the related pages.

The configuration function access menu is the following:



When a button is activated, it changes color from blue  (e.g.) to green  (e.g.), while its function displayed at the top of the screen  is activated to .

By touching / clicking on  or touching / clicking on again  (e.g.), the user accesses the corresponding page.

Refer to the example given below « *General configuration (Address, Protocol, Measurement channels...)* »:



NOTE : For memory, as explained in page 3-10, access to the configuration functions depends on the hierarchical level of the user’s password. There are three hierarchical levels: User, Advanced, and Expert.

The factory passwords available in the analyzer on delivery are the followings, symbolized by the icons:

Password level	Login	Icon
User	12345	
Advanced	78300	
Expert	00007	

On the initial start-up, it is strongly advised to change the factory passwords: activate  in the configuration functions menu to display the password management window, input the new passwords in the corresponding fields, and activate  to validate the input and close the password management window.

Configuration functions buttons description:

	Restore factory settings. System backup is recommended before performing this operation.
	Password management.
	General configuration. (Address, Protocol, Measurement channels...)
	Advanced analyzer configuration.
	Automatic cycles configuration.
	Optional interface configuration
	Find and display stored data.
	Delete all the stored data. (Warning! this operation can't be cancelled)
	Event history
	Diagnostic pages. (Internal parameters, input / output, measurement signals...)
	Configuration and software backup on USB key.
	Restore software and configuration from USB key.
	Record real time measurement data on USB key.
	Gives USB key characteristics (free and total memory) when connected to the analyzer.
	Leak test and pressure sensor calibration
	Automatic flow calibration
	Leak test
	O2 sensor calibration
	List of connected clients.

	Language selection.
	Local / Remote mode
	Set the analyzer to Maintenance mode.
	Put the analyzer on stand-by.

By touching / clicking on , the user returns to the home page.

DESCRIPTION OF ANALYZER FEATURES AND PAGES

STANDARD FUNCTIONS

Analyzer controls

Manual controls are used to manually drive the analyzer. There are two manual control types: those which select the gas to be measured (i.e. sample gas, zero gas, span 1, span 2...) and those which start automatic cycles (i.e. Zero Reference, Auto Span 1, Auto Span 2...).

The MIR 9000e has no internal solenoid valves but it can drive up to 8 external solenoid valves. The integrator or the user will have to set up the solenoid valves as needed, outside the analyzer, in the analysis system. The analyzer can directly power up to 6 solenoid valves (24V DC), enabling up to 3 span gas cylinders. If a 4th and a 5th span gas cylinder are required, the corresponding solenoid valves (SV7 and SV8 respectively) will need to be powered externally and driven using an optional input / output USB module.

The number of manual controls available depend on the external solenoid valves setup.

The SV1 (Zero SV), SV2 (Span SV) and SV3 (Span 1 SV) solenoid valves are always enabled by default. The SV4 to SV8 solenoid valves need to be declared in the « *Automatic cycle configuration* » page to be taken into account by the analyzer, and to display their corresponding manual controls.

1/ manual gas injection controls

Reminder: the icon located in the left corner at the base of the screen displays the analyzer status.

By default, when warm-up is over, the analyzer switches to Measurement mode, which is indicated by the  symbol.

Touching / clicking on the icon located to the left at the base of the screen displays the available manual gas injection controls. These controls will change the analyzer status.

At least and by default, the following controls are displayed:



When all the available solenoid valves are declared, all the manual controls are displayed:



When the user touches / clicks on one of these manual controls, the analyzer status changes and the symbol located to the left at the base of the screen is updated accordingly.

The manual gas injection controls are the followings:

Icon	Description
 Sample gas measurement	This control is used to manually switch the analyzer to sample gas measurement. This status can be interrupted at any time by triggering another zero gas or span injection control, or launching an automatic cycle. The corresponding analyzer status is "Measurement".
 Zero gas injection at analyzer level  Zero gas injection via the sampling probe	These controls are used to manually switch the analyzer to zero gas injection. The injection can be done directly at analyzer cabinet level, or upstream, at sampling probe level (if the SV probe SV5 is installed and declared in the « <i>Automatic cycle configuration</i> » page). The corresponding analyzer status is « Zero ». This operation is used to check the signal stability and drift. The comparison between the zero measured at analysis cabinet level and the zero measured via the sampling probe allows the detection of possible pollutions or leakages in the sampling system.
 Span gas n° 1 to 5 injection at analysis cabinet level  Span gas n° 1 to 5 injection via the sampling probe	These controls are used to manually switch the analyzer to the span gas n° 1 to 5 injection, depending on the solenoid valves installed and declared in the « <i>Automatic cycle configuration</i> » page. The injection can be done directly at analysis cabinet level, or upstream, at sampling probe level (if the SV probe SV5, is installed and declared in the « <i>Automatic cycle configuration</i> » page). The corresponding status is « Span i », with i = 1 to 5 depending on the chosen span gas. These operations are used to check the measurement stability and drift in response to span gases. The comparison between the measurements done at analysis cabinet level and the measurements done via the sampling probe allows the detection of possible pollutions or leakages in the sampling system. The user must use external calibrated span gases.

2/ Automatic cycles manual controls

As for the manual gas injection controls, the number of automatic cycles manual controls depends on the solenoid valves declared in the « *Automatic cycle configuration* » page.

Unlike the gas injection controls which only switch the analyzer from a status to another for an undefined length of time, the automatic cycle manual controls launch a cycle for a definite period at the end of which the analyzer recovers its initial status.

The cycles are divided in three steps:

- An initial purge, named Purge in. During this step, the gas contained in the measurement chamber is pumped out, and the gas corresponding to the launched cycle (zero or span) fills the measurement chamber.
- The main step, named Reference. During this step, the gas corresponding to the launched cycle (zero or span) is measured. If the cycle is a reference-zero or an auto-span, the internal analyzer parameters are recalculated (re-adjusted) at the end of this step.
- A second purge, named Purge out. During this step, the gas contained in the measurement chamber is pumped out, and the gas corresponding to the analyzer status before the cycle started (sample gas in general), fills the measurement chamber.
- The cycles are finished when the time displayed in the scroll bar reaches zero. The scroll bar is then entirely gray and closes automatically.

The duration of each step can be set up in the « *Automatic cycle configuration* » page.

Touch / click on  to display the automatic cycle manual controls.

- At least and by default, the following controls are displayed:



- When the SV4 (span 2 SV), SV6 (span 3 SV), SV7 (span 4 SV) and SV8 (span 5 SV) are declared (ON), the additional automatic cycles corresponding to these span gases are displayed :



- When solenoid valve SV5 (SV Probe) is not declared, the gas is only injected at analysis cabinet level.
- When the SV5 (SV probe) solenoid valve is present and declared (ON), the gases are injected either at analysis cabinet level or at sampling probe level. Two additional controls are displayed to select the gas injection channel :

⇒ Injection via the sampling probe:



Active status, inactive control



Inactive status, active control (switches to this status)

⇒ Direct injection at analysis cabinet level:

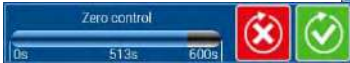
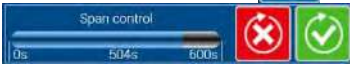


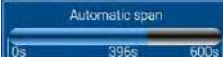
Active status, inactive control



Inactive status, active control (switches to this status)

The functions of the automatic cycle manual controls are the followings:

Icon	Description
 Zero check cycle	This control is used to manually launch a zero check cycle. It allows to check the zero gas response without re-adjusting the internal parameters. The user touches / clicks on  and views the cycle progress on the  scroll bar. At any time, it is possible to stop the current cycle by touching / clicking on  or .
 to  Span check n°1 to 5 cycles	These controls are used to manually launch a span n°1 to 5 check cycle, without re-adjusting the calibration factors. The user touches / clicks on  to  and views the cycle progress on the  scroll bar. At any time, it is possible to stop the current cycle by touching / clicking on  or . The user must use external calibrated gas cylinders.
 Ref. zero cycle	This control is used to manually launch an automatic zero-reference (zero re-adjustment) cycle. The user touches / clicks on  and views the cycle progress on the  scroll bar. At any time, it is possible to abort the current cycle by touching / clicking on , or validate and stop the current cycle by touching / clicking on . In both cases, the cycle stops and the analyzer recovers its initial status. If the cycle is validated, the zero is readjusted. If it is aborted, the zero is not re-adjusted.

Icons	Description
 to  Span check n°1 to 5 cycle	This control is used to manually launch an automatic span cycle. The user touches / clicks on  to  and views the cycle progress on the  scroll bar. At any time, it is possible to abort the current cycle by touching / clicking on , or validate and stop the current cycle by touching / clicking on . In both cases, the cycle stops and the analyzer recovers its initial status. If the cycle is validated, the calibration factor is readjusted. If it is aborted, the calibration factor is not re-adjusted. During the automatic span cycles, the analyzer automatically adjusts the "Span coeff." calibration factors so that the value read on the analyzer (minus the user offset) is equal to the span gas concentration value. The span gas concentration values are entered in the « <i>Advanced analyzer configuration</i> » page. The user must use external calibrated gas cylinders.

Measurement display

This is the measurement channel display. Column 1 gives the instant measurement values and column 2, the average measurement values.



To display additional measurement channels, it is necessary to first select them in the Measurement channel section of the « General configuration » page, as shown below (refer to green frame with arrow):



Buttons definition:

	Removes the selected measurement channel.
	Moves upwards the selected measurement channel.
	Moves downwards the selected measurement channel.
	Opens the selection pop-up window to add measurement channels.

To select additional channels, touch/click on  to open the measurement channel selection window.

There are five measurement channel groups: Compound, Voltage, Sensor, Calculation, and Analog inputs. By default, the compound group field is set to ON, the SO₂, NO and CO measurement channels are selected and displayed in the measurement channel section (refer to green frame with arrow in the above page). Up to 32 measurement channels can be selected simultaneously.



To select the other measurement channels, switch  to  in any other measurement channel group field to display the corresponding available channels, touch/click on the channels to be selected : they are displayed in white on blue sky background. Touch / click on  to validate this selection and close the window. The  key closes the window without validating.

It is possible to select all the available channels as shown below. Once they are selected, the channels disappear from right frame, thus preventing to select twice the same channel.

Details of the various measurement channel groups are shown below:

– Voltage measurement channel selection:



– Sensors measurement channel selection:



– Calculation measurement channel selection:



The selected measurement channels are placed in the numbered fields (refer to frame with arrow in the page below). Touch / click on  in the right side of the page to validate the whole selection.  is used to quit the page without validating.



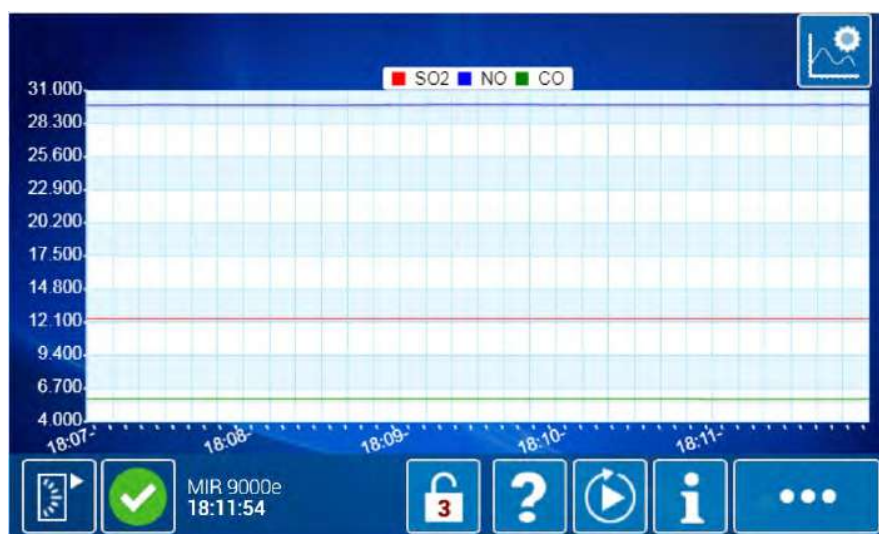
Touch / click on  to view the measurement channel display in the home page : only five measurement channels are displayed per page. To display all of the selected channels, the GUI (graphic user interface) adds as many pages as necessary. These additional pages are symbolized by the buttons  and  displayed at the top of the screen (refer to frame with arrow). Use these buttons to change page.



Real time graph display

Measurement channel values can be displayed as graphs. In order to do this, first select the desired channels by clicking on them, then touch / click on  and then on . If no channel is selected, an error message is displayed: "Select a measurement channel to display its corresponding curve ».

Up to 6 channels can be selected at the same time to be displayed as graphs.



By default, the X-scale setting is 5 minutes, and the Y-scale setting is automatic. The Y scale setting is adjusted in real time to allow each selected channel curve to be displayed on the same graph.



To configure the axis, touch / click on , switch AutoScale OFF, enter the desired values and confirm the changes with .



To activate the zoom function, the user selects the area to zoom by touching / clicking on it, or using the mouse pointer: it greys the selected area. The user validates the selection by touching / clicking on  to display the zoomed graph.



The user exits the zoomed graph by touching / clicking again on the screen.

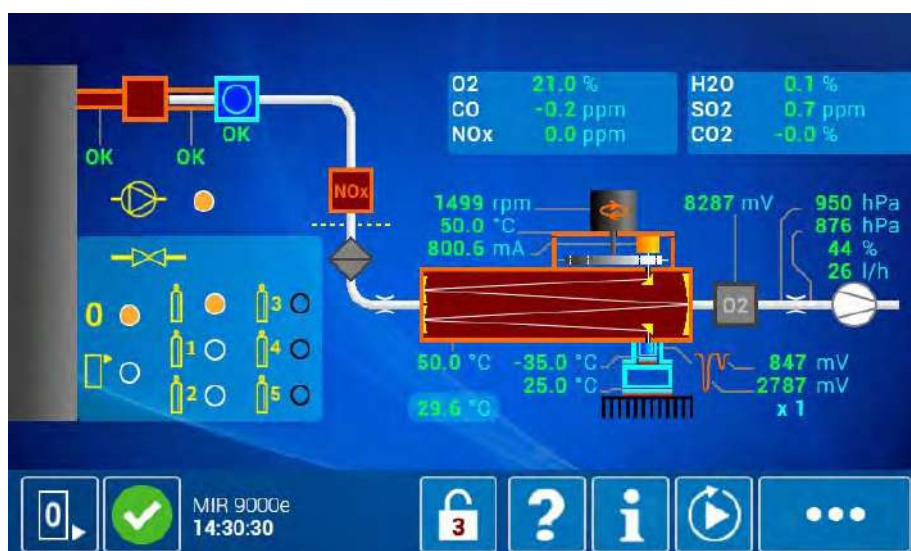
Synoptic visual display

The user accesses this page by touching / clicking on  and then on .

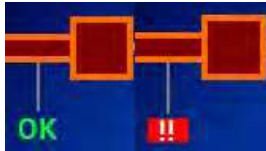
This page displays a synoptic diagram of the complete analysis system, including both external and internal components. External components are only shown if they are declared in the *Advanced Configuration* menu. Those which are not declared are only outlined, with no color filling. Numerical values are shown in green if they are within tolerances, or in white and red otherwise.

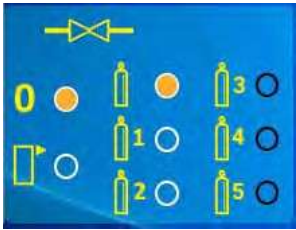
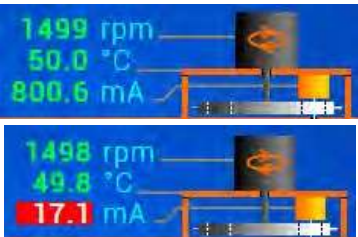
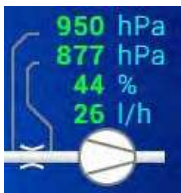
The yellow dashed line ----- symbolizes the analyzer boundary.

Measurement channels are also displayed on this page for convenience.



The different areas of the diagram are described in the table below. All the figures shown on the diagram are also displayed on the Diagnostic page.

Screen shot detail	Description
	Sampling tube and probe: OK means no alarm; !! means "Alarm"
	Heated line and gas cooler: Black outline means "not declared" Bold, color drawing means declared
	Heated line and gas cooler: OK means no alarm; !! means "Alarm"
	External NOx converter: Not declared, without alarm, and with alarm
	External sampling pump: OFF and ON

Screen shot detail	Description
	External solenoid valves Starting from the top and from left to right: Zero, Probe, Span, then span gas cylinders n°1 to n°5 Black circles: not declared White circles, blue inside: declared and idle White circles, orange inside: declared and powered
	Infrared source and correlation wheel assembly: Motor speed Temperature Infrared source current (without alarm / with alarm)
	Measurement chamber temperature
	Detector assembly. Left: infrared sensor temperature and detector assembly temperature. Right: lowest signal, highest signal, and amplifier gain
	Oxygen probe output signal
	Measurement chamber pressure Pump inlet pressure Pump power supply PWM duty cycle Flow rate

Information panel

The information panel is displayed by touching / clicking on  in the bottom browsing bar. It displays the software version, analyzer serial number, TCP/IP analyzer address, current analyzer date, data storage integration period, remaining days up to the next maintenance, if a USB device is connected to the analyzer, and number of connected clients.



Icons description:

	QR code
 icix	Shows that a WIFI dongle for the remote connection of a tablet or smartphone is fitted on the analyzer. ESA_MIR9000e_0 and 192.168.43.1 are WIFI TCP/IP identifier and address required for connection.
	Used to email analyzer status
	Used to email recorded events (history)
	Rings to locate the analyzer (can be useful when remotely operated).
	Closes the information panel.

Pop-up help operation

It is the same paragraph than page3-12.

The user activates the pop-up help by touching / clicking on  which turns green . Then, the user touches / clicks on button (e.g. ) or screen part to know the operation or use: a pop-up  opens to give the desired information. The pop-up help closes by touching / clicking on  .

NOTE : The pop-up help is disabled by touching / clicking on  again.

See the example below:



 button selected to know the operation or use,  message giving the desired information.

 button used to close the pop-up help.

CONFIGURATION FUNCTIONS

The user accesses the configuration function access screen by touching / clicking on .



The configuration functions can only be accessed by authorized users with a password. As indicated in page **Erreur ! Signet non défini.**, three hierarchical password levels are available: User, Advanced, and Expert.

All the configuration functions available for Expert users are described below.

Password management

The user touches / double-clicks on  to open this window.

The passwords indicated in the screenshot below are the factory passwords available in the analyzer on delivery. Upon initial startup, it is recommended to modify these factory passwords (refer to page **Erreur ! Signet non défini.**).

The user modifies the password fields by filling-in them and validating entry with .



closes the window without validating the modifications.



General configuration

The user accesses this page by touching / double clicking on . This icon remains on display in the top right corner of the screen.

This page allows the configuration of addresses and communication protocols, as well as measurement channels configuration.



: This button is displayed when the analyzer is connected to a computer, a smartphone or a tablet. By touching / clicking on it, the user updates the analyzer date and time with the current date and time of the connected equipment.

DATE and TIME fields: used to set the internal date and time of the analyzer.

ARCHIVE PERIOD field: used to set, in minutes, the periodicity of the stored average values (archives).

NETWORK section: used to configure the TCP-IP network.

- When the DHCP switch is ON, the analyzer network address is managed by a DHCP server.
- When the DHCP switch is OFF, the analyzer ADDRESS, MASK and GATEWAY can be entered by the user.

UDP SERVER section: used to configure the two available UDP ports, Port A and Port B.

- PORT A and PORT B: the port number is programmable from 1000 to 9999.
- ADDRESS: these are only used for Mode4 (four alphanumeric characters) and JBUS (number between 0000 and 0255). Default setting : MIR2
- PROTOCOL combo box: Mode 4, PRN, and JBUS can be used.

TCP/IP MODBUS SERVER section: used to configure MODBUS TCP communication.



The screenshot shows the configuration interface of the MIR 9000e analyzer. The screen has a dark blue background with white text and various input fields. At the top, there are fields for Date (2020-04-01), Time (16:40:55), and Archive period (1). Below this is the Network section, which includes a DHCP toggle switch (ON) and a WPA key field (123456789). The UDP server section has two rows for Port A and Port B, each with an Address field (MIR2 and 0001) and a Protocol dropdown menu (Mode4 and JBUS). The TCP/IP Modbus server section has an Active toggle switch (ON) and an Address field (502). The Operation section has a Custom ident. field and an Automatic lock (sec) field (14400). On the right side, there is a vertical bar with a gear icon at the top, a green checkmark button, and a red back arrow button. At the bottom, there is a status bar with a lock icon, a question mark icon, an information icon, a play icon, and a home icon. The status bar also displays the device name (MIR 9000e) and the current time (16:41:43).

OPERATION section:

- CUSTOM IDENT.: allows the user to define a custom identification name or number
- AUTOMATIC LOCK (SEC): used to define the time after which the HMI gets locked up and asks the user to enter a password again. Unit: seconds.
- NEG. VALUES: if this switch is ON, negative values are displayed. If it is OFF, negative values are replaced by null values.
- SECURITY: if this switch is ON, the analyzer switches automatically to safety mode when potentially damaging faults are detected. The analyzer safety mode consists of the following steps:
 - Firstly, the sampling system is purged with zero gas
 - Afterwards, the analyzer is put on stand-by and the pumps are stopped.

It is strongly recommended to leave this switch ON in order to protect the analyzer in case of failure of a critical system component (for example: if the sampling probe temperature is too low or if the residual humidity is too high).

- ALARM: this switch enables or disables alarms management.

MEASUREMENT CHANNEL section: used to configure measurement channels (refer to § 3.4.1.2).



Advanced analyzer configuration

Touch or click on  to enter these pages. These pages are used to configure the metrological parameters, the general operation of the analyzer and the external components which are connected to it.

Page 1:

COMPOUNDS sections:

- UNIT: selects the compound measurement unit. Possible units are %, mg/Nm³, and ppm.
- CONV. COEFF: displays the factor used to convert ppm to mg/Nm³. This factor can be changed if desired.
- SPAN COEFF. and OFFSET: these two values are used to apply an $Y = Ax + B$ calibration curve to the measurement. A is the calibration factor and B is the offset. The calibration factor can either be entered by the user or modified during an automatic self-calibration cycle.
- DELTA (%): when the calibration factor is modified by an automatic cycle, this value indicates the difference between the new and the previous factor. If this value is greater than 50%, a span alarm is triggered, the analyzer ignores the new calibration factor and keeps the previous one. It is then necessary to manually re-validate the new calibration factor to disable the alarm.
- RANGE: allowed measurement range, in the unit specified in the UNIT field. If the measurement value exceeds the range value, a range overflow alarm is triggered.
- SPAN GAS VALUE: span gas cylinder concentration used to perform calibration checks and automatic self-calibration, expressed as ppm or %.
- CALIB.: used to allocate the compound to an automatic self-calibration cycle, and thus to a span cylinder. When the allocated cycle runs, the compound's calibration factor is re-adjusted. Only one cycle can be allocated to a given compound, but several compounds can be allocated to a given cycle (for example if a mixed span gas cylinder is used).
 - OFF disables the automatic self-calibration for the compound.
 - At least and by default, when only the SV1 (SV zero), SV2 (SV span) and SV3 (SV span 1) are declared in the "Automatic cycle configuration" page, only the "Span 1" auto-span cycle is available.
 - When the solenoid valves corresponding to the span cylinders n°2, 3, 4 and 5 are declared in the "Automatic cycle configuration" page, the Span 2, 3, 4 and 5 automatic self-calibration cycles become available.
 - When the SV5 (SV probe) solenoid valve is declared, the available auto-span cycles are doubled. The chosen cycles can inject gas either in the analysis cabinet (example: auto span.1), or in the sampling probe (example: auto span^1).



The screenshot displays the configuration interface for three compounds: NOx, SO2, and CO. Each compound has a set of parameters including Unit, Conv. coeff, Span coeff, Delta (%), Offset, Range, Span gas value, and Calib. The interface includes a top navigation bar with icons for home, settings, and help. A bottom status bar shows the device name 'MIR 9000e', the time '09:17:51', and various status icons like a lock, question mark, information, and home button.

Compound	Unit	Conv. coeff	Span coeff	Delta (%)	Offset	Range	Span gas value	Calib.
Compound NOx	mg/Nm ³	2.05	1.023	1.02	0	200	88 ppm	Auto span.1
Compound SO2	mg/Nm ³	2.86	0.986	0.99	0	300	79 ppm	Auto span.2
Compound CO	ppm		1	0.00	0	2500	88 ppm	Auto span.2

Page 2:

This page is used to access the pressure and flow rate measurement calibration parameters, and other analyzer operation parameters.

PRESSURE AND FLOW CALIBRATION section:

The A and B values are respectively the calibration factors and offsets of the $Y = Ax + B$ calibration curves used for internal pressure and flow rate measurements. Pressure is measured at two places: at the measurement chamber (*Bench pressure*) and at the pump inlet (*Pump pressure*). The flow rate is calculated using the difference between these two pressure measurements. The flow rate offset is normally 0.

To calibrate the pressure and flow rate measurements, it is recommended to use the following semi-automatic functions, accessible from the main menu:



Leak test and pressure sensor calibration (see § 3.4.2.9)



Automatic flow rate calibration (see § 3.4.2.10)

Alternatively, the A and B values can also be entered manually.

The measured values are shown in orange characters on the right hand side of the B value boxes.

GLOBAL PARAMETERS section:

- DILUTION ON/OFF switch:
 - If ON, a « COEFF. » field is displayed to enter the dilution factor.

- PUMP STATUS: this combo box is used to drive the analyzer pump:
 - OFF: stops the pump
 - ON: starts the pump. The pump power supply is servo controlled by the measurement chamber pressure. SETPOINT is used to enter the measurement chamber pressure set point. This is the normal operating mode of the analyzer.

The default set point value is 950 hPa. This set point may need to be lower if the analyzer is operated at altitude, or without an external primary sampling pump (which is not recommended).

The percentage shown in orange characters to the right hand side of « Auto. » is the duty cycle of the pump PWM power supply. During normal operation, this percentage should be between 35% and 60% ; ideally between 40% and 55%. If the duty cycle drops below 35%, the pressure set point should be lower. If the duty cycle exceeds 60%, the pressure setpoint should be higher. In both cases, one should ensure that the sample gas is correctly supplied to the analyzer inlet, without any excessive pressure drop or overpressure.

- FORCED: powers the pump at a constant PWM duty cycle which must then be specified in the CYCLE box. The dynamic range of this number is approximately between 35% and 60%. This mode may be useful to perform tests, but is not recommended in normal operation.

- FILTERWHEEL ON/OFF switch: starts or stops the filterwheel motor. This switch must be used to restart the filterwheel motor if the latter has been stopped for security reasons, for example following the detection of an abnormally low speed due to friction or jamming. Proceed carefully in this case; technical investigation may be required to identify the cause of the problem.
- ZREF. LATCH ON/OFF switch :
 - OFF: the analyzer keeps measuring as normal throughout Zero Reference cycles. Thus, measurement values drop to zero momentarily.
 - ON: the analyzer latches the measurement values during Zero Reference cycles. Thus, the measurement values remain constant. They stay equal to those obtained just before the Zero Reference cycles started.
- NORMALIZATION: REFERENCE VALUES section: each measurement may be normalized as a function of H₂O, O₂ and CO₂ concentration (all expressed as %).
 - The normalization calculations are as follows :
 - H₂O normalization: the reference must be equal to 0. The formula is:
 - $\text{Meas. Value (normalized)} = \text{Meas. Value} \times 100 / (100 - (\text{H}_2\text{O conc.}))$
 - O₂ normalization: the reference must be specified. The formula is:
 - $\text{Meas. Value (normalized)} = \text{Meas. Value} \times (21 - (\text{O}_2 \text{ Ref.})) / (21 - (\text{O}_2 \text{ conc.}))$
 - CO₂ normalization : the reference must be specified. The formula is:
 - $\text{Meas. Value (normalized)} = \text{Meas. Value} \times (\text{CO}_2 \text{ Ref.}) / (\text{CO}_2 \text{ conc.})$

The screenshot displays the MIR 9000e control interface with the following sections:

- Pressure and Flow measurement calibration:**

	A	B	
Bench pr.	1.0192	-7.12231	950 hPa
Pump pr.	1.01865	-8.88062	863 hPa
Flow	1.01811	0	26.7 l/h
- Global parameters:**
 - Pump status: ON
 - Setpoint: 950 hpa
 - Cycle: Auto. 46 %
 - FilterWheel: ON
 - ZRef. latch: OFF
 - Dilution: OFF
- Normalization : reference values:**
 - H2O (%): OFF 0
 - O2 (%): OFF 11
 - CO2 (%): OFF 11

The bottom status bar shows: MIR 9000e, 09:21:07, and navigation icons for calibration, status, help, information, play, and home.

Page 3:

This page is used to declare the external components connected to the analyzer.

EXTERNAL ELEMENTS section:

The sampling probe, the heated line, the dryer and the NOx converter can all pass alarm signals to the analyzer, for example using alarm relays and the analyzer's optional input / output USB modules.

The ON/OFF switches in the column named PRESENT are used to declare the connected components so that they may be displayed on the synoptic visual display page (cf §3.4.1.4).

When an external component is declared, its alarm signals may be taken into account by the analyzer by selecting YES in the ALARM FEEDBACK combo box. Their status (OK or Alarm) will thus be displayed on the synoptic visual display page and also be used to disable measurements and put the analyzer into security mode when faults happen.

The sampling probe is declared PRESENT by default.

The NOx converter can belong to two different types:

- Either it is independent from the analyzer, for example if it is made by a third party manufacturer. In this case, it is not driven by the analyzer, and the STATUS combo box must be set to EXTERNAL.
- Or it is driven by the analyzer, which is the case with our dedicated MIR 9000e NOx converter option. In this case, its status can be ON or OFF and must be selected in the STATUS combo box. The NOx converter default temperature set point is 290°C. The measured temperature is displayed in orange characters on the right hand side of the SETPOINT value. If the option is not connected, that value should be ignored.

When an external primary sampling pump is connected to the analyzer, the latter can be started or stopped using the ON/OFF switch.

O2 PROBE section:

The first combo box is used to declare the presence or absence of the O2 probe and define whether it is internal or external to the analyzer. If it is an external one, it must be connected to the analog input located on the rear panel of the analyzer (see §1.1.2.2).

When an internal O2 probe is declared, the second combo-box is used to specify its type, which can be either Zirconium or Paramagnetic.

AUX combo box:

This combo box is used to allocate the AUX connector to either an external pump or to any declared solenoid valve, including those already driven through the 10 points connector (see §1.1.2.2).

Example : if the analyzer does not drive a primary pump, but if 3 span gas cylinders must be operated, the AUX connector should be set to SV6 (for SPAN 3) so that the analyzer can drive directly all the required solenoid valves.

External elements	Present	Alarm feedback	Status	Setpoint
Sample probe	Yes	Yes		
Heated line	ON	No		
Dryer	ON	Yes		
NOx converter	ON	No	OFF	290 448.0 °C
External pump	ON			

O2 probe: Internal Zirconium

Aux: External pump

MIR 9000e 10:37:13

Automatic cycle configuration

The user accesses this page by touching / double-clicking on . This icon remains displayed in the top right corner of the screen.

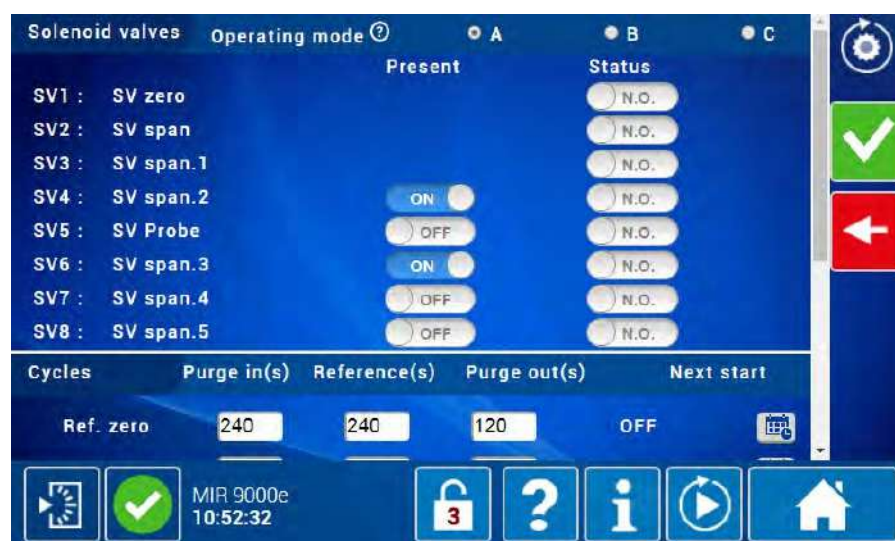
This page is used to configure the analysis system solenoid valves and the analyzer automatic cycles correlated with the solenoid valves management. These cycles can be launched either manually or automatically according to a predefined schedule. (Refer to section 3.2.1.1 *Analyzer controls*)

SOLENOID VALVES section:

This section is used to declare the solenoid valves present in the analysis system and to configure their operating modes according to the analysis system design type (A, B or C) and their operating type (N.O. = normally open, N.C. = Normally closed).

The supported system design types are detailed in Chapter 2 (refer to figures 2-2 to 2-6).

- OPERATING MODE: touch / click on the A, B or C buttons to select one of the three possible solenoid valves operating modes. The simplified drawings of the corresponding system designs are displayed by touching / clicking on  which becomes , then by touching / clicking on A, B or C.
- SV1 to SV8: declaration of the analysis system solenoid valves.
- PRESENT switches: SV1 (SV zero), SV2 (SV span) and SV3 (SV span 1) are enabled by default. The other solenoid valves (SV4 to SV8) can be declared present (ON) or absent (OFF).
- STATUS switches: used to configure the solenoid valves operating type (N.O. = normally open, N.C. = normally closed).



CYCLES section:

The available cycles depend on the declared solenoid valves. At least four cycles are always available:

- REF. ZERO cycle: injects zero gas and readjusts the analyzer zero.
- ZERO CHECK cycle: injects zero gas to check the analyzer response, without readjusting the analyzer zero.
- SPAN CHECK 1 cycle: injects span gas from cylinder n°1 to check the analyzer response, without readjusting the analyzer calibration factor.
- AUTO SPAN 1 cycle: injects span gas from cylinder n°1 to perform an automatic self-calibration. The calibration factors of the compounds allocated to cylinder n°1 (refer to “Advanced analyzer configuration” menu) will be automatically readjusted.

When the SV4, SV6, SV7 and SV8 solenoid valves are declared present (ON), new cycles are displayed: they correspond to the span gas cylinders n°2, 3, 4, and 5. These are the same cycles as SPAN CHECK 1 and AUTO SPAN 1 described above but with the other span gas cylinders.

When SV5 (SV probe) is declared present (ON), the cycles previously described are doubled. From now on, each gas injection can be done either in the analysis cabinet or in the sampling probe:

- The cycles named as previously correspond to gas injections taking place in the analysis cabinet
- The cycles whose names finish with ^ correspond to gas injections taking place in the sampling probe

Description of the CYCLES section:

- PURGE IN (S): sets the initial purge duration. During this step, the gas contained in the measurement chamber is pumped out, and the gas corresponding to the launched cycle (zero or span) fills the measurement chamber.
- REFERENCE(S): sets the duration of the cycle's reference measurement. During this step, the gas corresponding to the cycle (zero or span) is measured. If the cycle is a reference-zero or an auto-span, the internal analyzer parameters are re-adjusted at the end of this step.
- PURGE OUT(S): sets the final purge duration. During this step, the gas contained in the measurement chamber is pumped out, and the gas corresponding to the analyzer status before the cycle started (sample gas in general), fills the measurement chamber.
- NEXT START: shows the date and time of the next scheduled cycle. The  button is used to schedule the automatic cycles. OFF shows that no automatic cycle is scheduled.

The purge durations need to be longer than the total response time of the system. In general, the purge time is greater when gas is injected in the sampling probe than when it is injected in the analysis cabinet because of the extra time it takes to the gas to flow back and forth between the analysis cabinet and the sampling probe.

In order to optimize measurement accuracy, purge time should be about 4 times longer than the time required by the system to measure 90% of a step change in gas concentration value. This is important in particular for the PURGE IN step which will determine the zero and span accuracy up to the next cycle.

ZERO REF. AT STARTUP: this function is optional. This switch is used to force the analyzer to do a Zero-Ref cycle just after warm-up.

Cycles	Purge in(s)	Reference(s)	Purge out(s)	Next start
Ref. zero	240	240	120	OFF
Zero check	240	240	120	OFF
Span check.1	240	240	120	OFF
Span check.2	240	240	120	OFF
Span check.3	240	240	120	OFF
Auto span.1	240	240	120	OFF
Auto span.2	240	240	120	OFF
Auto span.3	240	240	120	OFF

Zero Ref. at startup OFF

✓

MIR 9000e
10:53:44

3

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🏠

Cycle planning

Span check.3

Periodicity ON

Period(s) 7 Days

Next start 2020-04-05 08:00

✓

✗

Find and display stored data

The user accesses this page by touching / double-clicking on . This icon remains displayed at the top right corner of the page.

When this page is open, it displays a pop-up menu used to set the date and time of the beginning and the end of the stored data period the user wants to see.

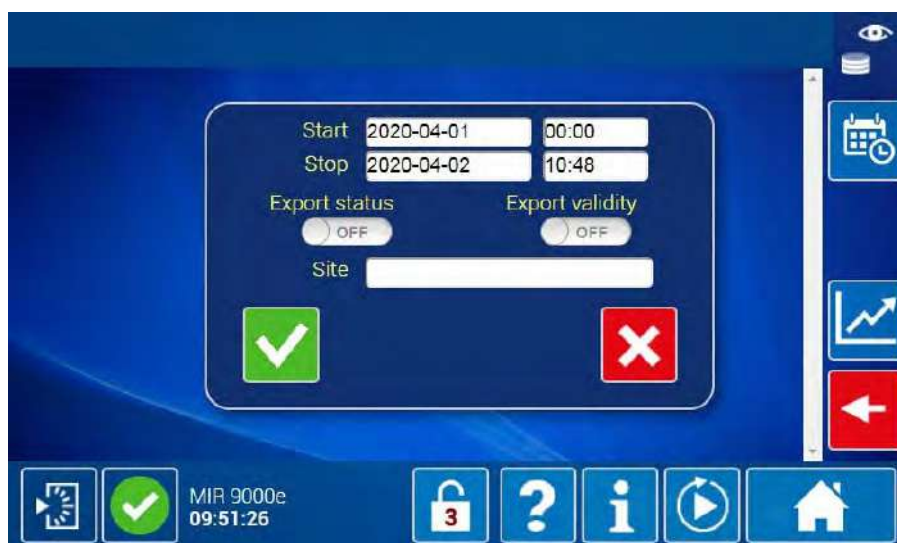
Touch/click on the Start and Stop fields to enter the date and time of the desired period, then on  to validate the choice, close the pop-up window and display the data. To simply close the pop-up window, touch / click on .

EXPORT STATUS ON/OFF switch: when a text file is generated, the status code of the analyzer can be either exported or not. The status code is a bit state sum defined as follows:

BIN	HEX	DEC	Description
00000000	00	0	Valid measurement
00000001	01	1	Invalid measurement (example : stand-by mode)
00000010	02	2	General alarm
00000100	04	4	Calibration
00001000	08	8	Zero mode
00010000	10	16	Span mode
00100000	20	32	Maintenance Mode
01000000	40	64	Less than 2/3 of the stored element duration correspond to the indicated status code (typically when the analyzer status changes)
10000000	80	128	Warm-up
11111111	FF	255	Configuration change

EXPORT STATUS ON/OFF and EXPORT VALIDITY ON/OFF switches: when a text file is generated, the validity percentage of the measurement intervals can be either exported or not.

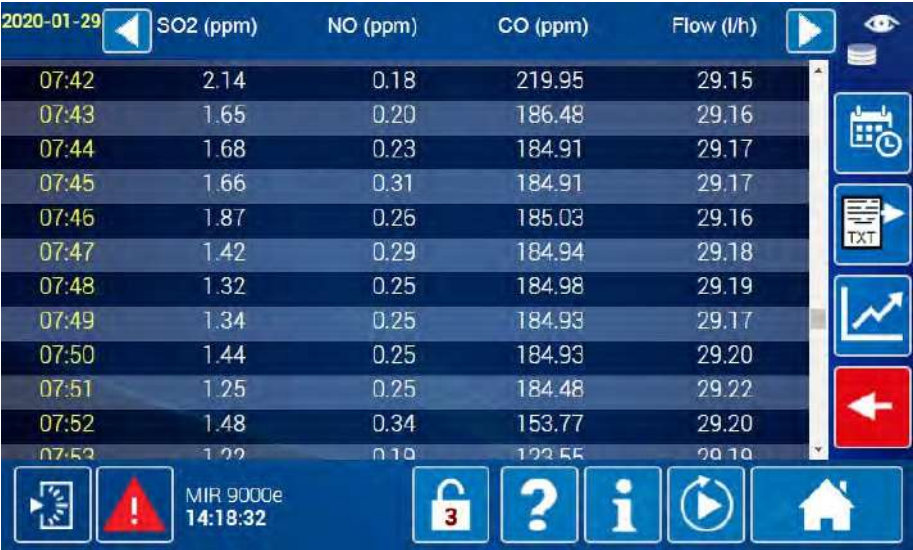
SITE field: used to enter a site name in order to help with data management. This field is useful when exporting data as text files.



The stored data is displayed as a table.

The columns correlate with the displayed measurement channels, i.e. those declared in the “*General configuration*” menu. In fact, all the data is recorded, whether it is displayed or not. To access those data, the only thing to do is to go to the “*General configuration*” menu and add it to the list of the displayed measurement channels, before returning to the present page.

The lines correlate with the data values averaged during each « *Archive period* » time interval. This time interval is set in minutes in the « *General configuration* » menu.



The  and  buttons are used to shift through the table columns.

Buttons description:

	Reopens the pop-up menu in order to change the date and time of the data to be displayed.
	Exports the displayed data in a TXT file, when the HMI is open via a computer.
	Displays graphs of selected measurement channels. Select up to 6 channels by clicking on their headers.
	Quits the page to return to the advanced functions home page.

Meaning of the data table icons

Icons may be displayed in the data table:



Their meaning is the following:

	This icon shows an invalid data line with an “alarm” status. Alarm details can be displayed by touching / clicking on the icon.
	This icon shows an invalid data line. Details of the analyzer status can be displayed by touching / clicking on the icon.
	This icon shows that a calibration event occurred. This event can be either a zero or span check, a reference-zero or an auto span. Details of this event can be displayed by touching / clicking on the icon.

Delete all stored data

The user deletes all the recorded averages by touching / double-clicking on  then on 

WARNING: this operation can’t be cancelled.

History

The user accesses these pages by touching / double-clicking on .

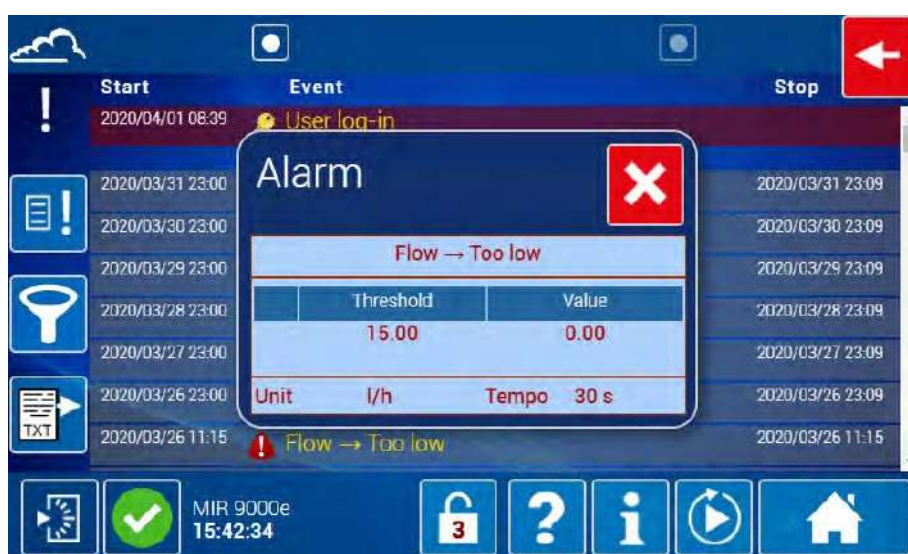
These pages are used to view all the events stored in the analyzer since the last stored data deletion.

Page 1:

- START column : displays the date and time of the beginning of the event
- STOP column: displays the date and time of end of the event
- EVENT column: displays the type and location of the event, and qualifies it with the following symbols.

	SIMPLE EVENT: the analyzer operates normally.
	CONFIGURATION MODIFICATION
	ALARM: the analyzer or the sampling system operation is faulty.
	SECURITY: alarm which could damage the analyzer. The analyzer switches to stand-by mode for safety.

The user can open a pop-up window giving the event details by touching / clicking on the icon. In case of alarm, the window displays the threshold values and the unit, the alarm delay, and the parameter value which triggered the alarm.



Buttons definition

	This button is used to display or hide the stored (past) events.
	This button is used to display only some types of events. Pressing several times hides and displays different types of events.
	This button is used to export the displayed data into a .TXT file.

Page 2:

This page is used to display the history of the modifications which occurred to the following parameters: « Span coefficient », « Zero reference », « Normalization », « Conversion coefficient », « Span gas value » and « Offset ». It makes it easy to audit the analyzer by accessing the traceability of all the user accessible parameters which affect the measurement results.

Touch / click on the calendar icon  to display the pop-up window.

The START and END fields are used to select the period dates and time of the history to be displayed.



Diagnostic features

The user accesses this pages by touching / double-clicking on .

The first page includes 6 sections: « Power supply and Boards », « Optical bench », « Fluid circuits », « Fans », « NOx converter » and « O2 probe ». Each section is open or closed by touching / clicking on its heading.



 is used to quit the page and return to the advanced functions home page.

Buttons and symbols description:

	Displays / hides the second decimal place
	This picture represents the inputs/outputs (for illustration only)
	This picture represents a voltmeter (for illustration only)
	This status icon indicates that the displayed values are within their tolerances

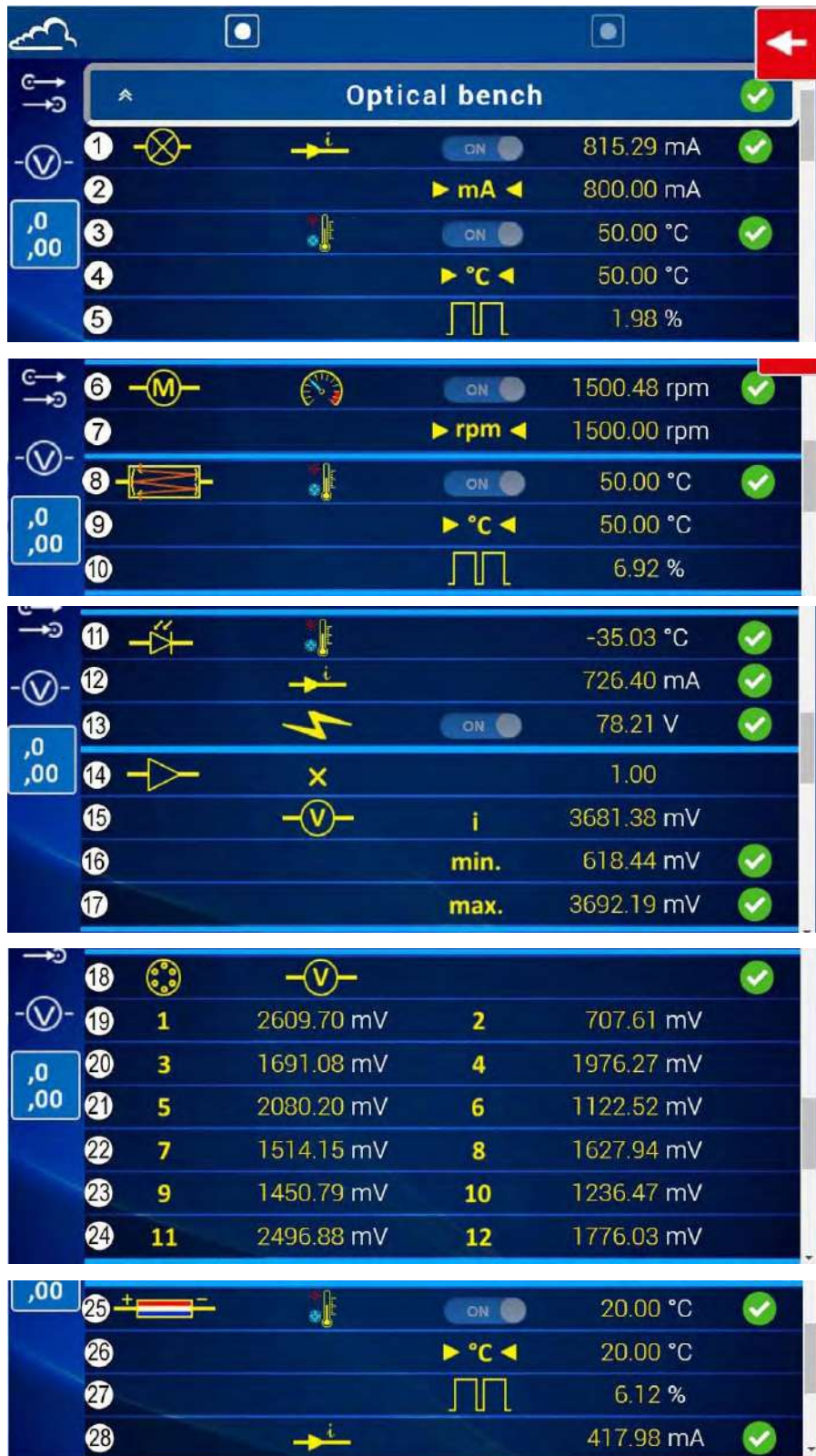
POWER SUPPLY AND BOARDS section:

Line Nb	Description
1	24V secondary power supply: ON/OFF status, measured current and voltage
2	12 V power supply: measured voltage
3	9 V power supply: measured voltage
4	5 V power supply: measured voltage
5	4 V reference: measured voltage
6	- 4 V reference: measured voltage
7	3,3 V power supply: measured voltage
8	5 V sensor power supply: ON/OFF status and measured voltage
9	Measurement board health state (MX microcontroller) & connection status
10	Measurement board health state (MZ microcontroller) & connection status
11	Controller board health state & connection status



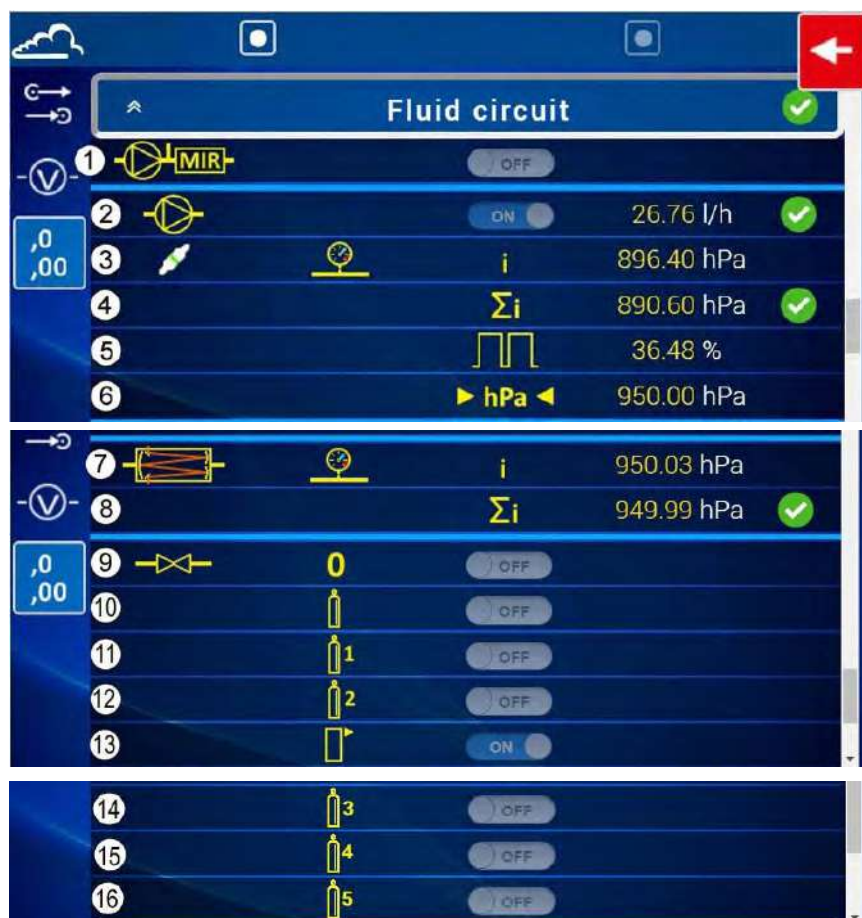
OPTICAL BENCH section:

Line Nb	Description
1	Infrared lamp power supply: ON/OFF status and measured current
2	Infrared lamp power supply: current setpoint
3	Source assembly temperature regulation: ON/OFF status and measured temperature
4	Source assembly temperature regulation: temperature setpoint
5	Source assembly temperature regulation: PWM power supply duty cycle = used power / available power
6	Correlation wheel motor: ON/OFF status and measured rotation speed
7	Correlation wheel motor: rotation speed setpoint
8	Measurement chamber temperature regulation : ON/OFF status and measured temperature
9	Measurement chamber temperature regulation : temperature setpoint
10	Measurement chamber temperature regulation : PWM power supply duty cycle = used power / available power
11	Detector: internal temperature
12	Detector: internal Peltier cooler current consumption
13	Detector polarization voltage: ON/OFF status and measured value
14	Programmable gain amplifier gain (normal setting = 1)
15	Instant detector signal
16	Minimum signal envelope value
17	Maximum signal envelope value
18 à 24	Measurement signal values for the 12 positions of the correlation wheel
25	Detector assembly temperature regulation: status (ON/OFF) and temperature measured
26	Detector assembly temperature regulation: temperature setpoint
27	Detector assembly temperature regulation: PWM power supply duty cycle
28	Detector assembly Peltier cooler current consumption



FLUID CIRCUIT section:

Line Nb	Description
1	Optional external sampling pump status (ON/OFF)
2	Internal analyzer sampling pump ON/OFF status and analyzer flow rate
3	Analyzer pump inlet pressure: instant value
4	Analyzer pump inlet pressure: filtered value
5	Analyzer pump power supply: PWM power supply duty cycle
6	Analyzer pump management mode and setpoint. Default setting: measurement chamber pressure setpoint
7	Measurement chamber pressure: instant values
8	Measurement chamber pressure: filtered value
9	SV1 (SV Zero) solenoid valve status
10	SV2 (SV Span) solenoid valve status
11	SV3 (SV Span 1) solenoid valve status
12	SV4 (SV Span 2) solenoid valve status
13	SV5 (SV Probe) solenoid valve status
14	SV6 (SV Span 3) solenoid valve status
15	SV7 (SV Span 4) solenoid valve status
16	SV8 (SV Span 5) solenoid valve status



FAN section:

Line Nb	Description
1	Analyzer internal temperature
2	Detector assembly heatsink fan status (ON/OFF)
3	Analyzer air extraction fan status (ON/OFF)

NOx CONVERTER section:

Line Nb	Description
1	ON/OFF status and temperature of the optional external NOx converter connected to the analyzer. Ignore this value if no external NOx converter is connected.
2	Temperature setpoint of the optional external NOx converter connected to the analyzer.
3	Temperature regulation of the optional external NOx converter: PWM power supply duty cycle = used power / available power

SONDE O2 PROBE section:

Line Nb	Description
4	Internal O2 probe analog signal: instant value
5	Internal O2 probe analog signal: filtered value



The second page, identified by the  symbol, displays the intermediate calculation variables used to compute gas concentration. The calculation variables are described in the tables below.

Compounds measured by the infrared measurement module:

Variable	Description
S	Compound measurement signal (at the absorbed wavelengths)
R	Compound reference signal (at the non-absorbed wavelengths)
S/R	S/R Ratio
S0	Compound measurement signal during the last Zero-Reference
R0	Compound reference signal during the last Zero-Reference
S0/R0	S0/R0 Ratio
Abs	Measured (uncorrected) absorption
Correction	Correction applied to the absorption value (related to cross-sensitivities between different compounds)
Corr. Abs.	Corrected absorption = Measured absorption - Correction
Conc.1	First step of concentration calculation. The linearization function is applied to the corrected absorption. Unit = ppm.
Conc.2	Second step of concentration calculation. The gas mixture correction is applied. Unit = ppm.
Conc.3	Third step of concentration calculation. The final value is obtained after applying a noise filtering algorithm, pressure corrections, the dilution factor, the calibration factor, the user offset, normalization calculations, and finally, unit conversion into mg/Nm3 or % vol.



SO2			
S	2243.681	S0	2241.357
R	2128.458	R0	2125.642
S/R	1.0543	S0/R0	1.0544
Abs	0.0140 %	Correction	0.0014 %
Corr. Abs.	0.0126 %	Conc.1	0.74 ppm
Conc.2	0.74 ppm	Conc.3	0.46 ppm
CO			
S	2776.295	S0	2774.129
R	1870.858	R0	1868.181

Oxygen measurement module:

Variable	Description
Inst. Sig.	Instant signal
Signal	Filtered signal
Pr. Part.	Partial oxygen pressure
Conc.1	Calibration function application. Unit = ppm.
Conc.2	= Conc.1
Conc.3	Final value obtained after applying a noise filtering algorithm, pressure correction, dilution factor, calibration factor, user offset, normalization calculations, and finally, conversion into % vol.

O2			
Inst.sig.	8300.45 mV	Signal	8300.55 mV
Pr. Part.	200.12 mbar	Conc..1	2.11e+5 %
Conc..2	2.11e+5 %	Conc..3	21.07 %

Final Summary Table:

This table lists the following values for each compounds:

- Measured (uncorrected) absorption
- Correction applied to absorption
- Corrected absorption
- Concentration 1
- Concentration 3

Compound	Raw Abs	Correction	Corr. Abs	Conc. 1	Conc. 3
NOx	0.037	0.034	0.003	0.059	0.054
SO2	0.035	0.003	0.032	1.874	1.321
CO	0.036	0.000	0.036	0.357	0.370
CO2	-0.065	0.000	-0.065	-470.795	-0.051
H2O	-0.108	-0.002	-0.105	-575.708	-0.059
NO	0.037	0.033	0.003	0.076	0.073

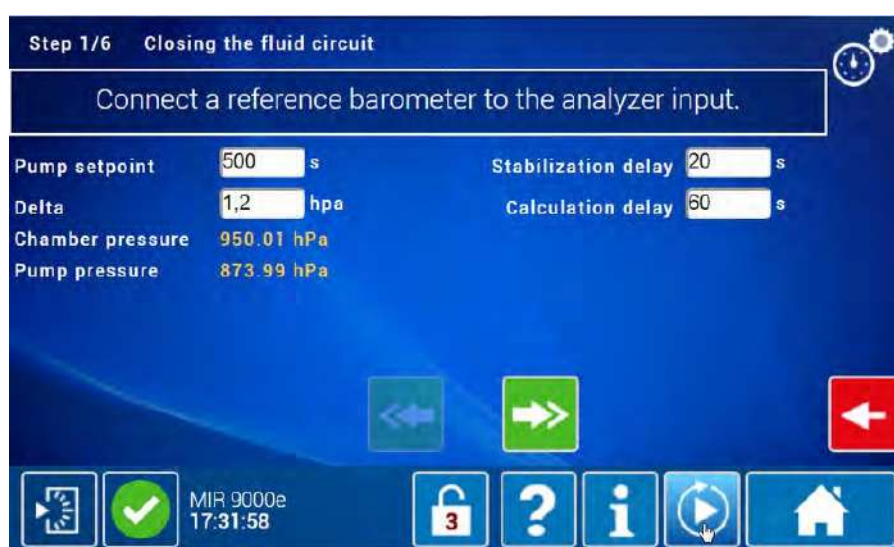
Leak test and pressure sensor calibration

Touch / double-click on  to access this function.

The procedure comprises 6 successive steps. A wizard guides the user step by step to carry out the leak test and the pressure sensors calibration. When prompted, after having done the required actions, touch / click on  to move from one step to the next. Alternatively, touch / click on  to move back to the previous step, or on  to abort the procedure.

The procedure requires the following tools:

- A reference barometer for absolute pressure measurement. The barometer must be able to measure up to atmospheric pressure at least, with a 0.1 hPa accuracy.
- A clamp in order to pinch the soft tubes inside the analyzer.



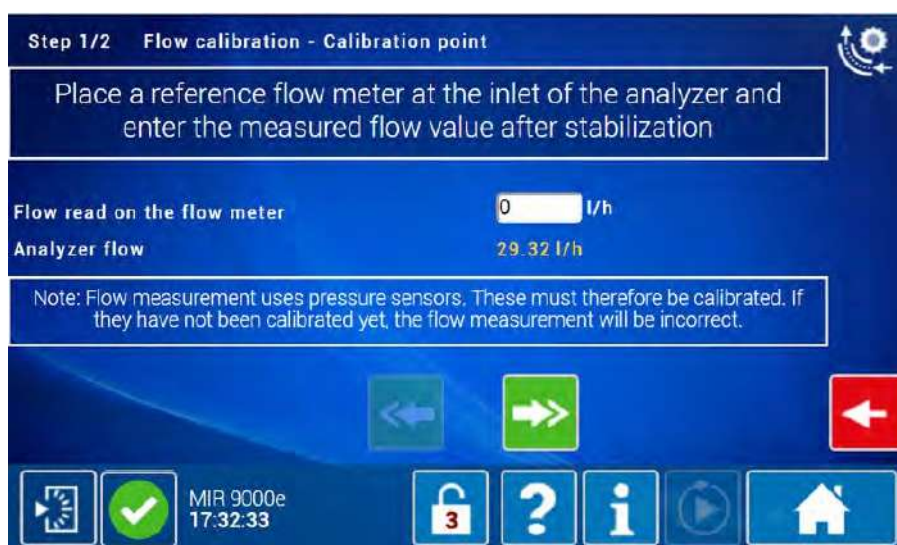
Automatic flow calibration

Touch / double-click on  to access this function.

The procedure comprises 2 successive steps. A wizard guides the user step by step to carry out the leak test and the pressure sensors calibration. When prompted, after having done the required actions, touch / click on  to move from one step to the next. Alternatively, touch / click on  to move back to the previous step, or on  to abort the procedure.

A reference flow meter, such as a rotameter, is required for this procedure.

NOTE : The flow measurement uses the pressure sensors. These must therefore be calibrated. If they have not been calibrated yet, the flow measurement will be incorrect.



Leak test

Touch / double-click on  to access this function.

The procedure comprises 2 successive steps. A wizard guides the user step by step to carry out the leak test and the pressure sensors calibration. When prompted, after having done the required actions, touch / click on  to move from one step to the next. Alternatively, touch / click on  to move back to the previous step, or on  to abort the procedure.

The procedure requires the following tools:

- A plug for the sample gas inlet;
- A clamp in order to pinch the soft tubes inside the analyzer.



O2 sensor calibration

Touch / double-click on  to access this function.

The procedure comprises 4 successive steps. A wizard guides the user step by step to carry out the leak test and the pressure sensors calibration. When prompted, after having done the required actions, touch / click on  to move from one step to the next. Alternatively, touch / click on  to move back to the previous step, or on  to abort the procedure.

The procedure requires dry instrument air.

If calibrating a Zirconium module, this procedure is sufficient.

If calibrating a Paramagnetic module for the first time, the potentiometers on the analyzer rear panel must be adjusted first. Refer to § 3.5.5 for that procedure.

NOTE : This procedure adjusts the O2 module analog signal transfer function. It does not affect the user calibration factor accessible from the Advanced Configuration page. The user calibration factor can be adjusted either manually, or by running a self-calibration (auto-span) cycle.



List of connected clients

Touching / double-clicking on  opens a pop-up window which displays the IP addresses and inactivity duration of the connected clients.

 closes the window.



Language selection

Touching / double-clicking on  opens a pop-up window which can be used to change the analyzer language.

The red frame around the flag indicates the language in use. The user touches/clicks on the desired flag to choose a language and close the window.



Local / Remote

Double touch / click on  to switch the analyzer into Remote mode.  then becomes .

When the Remote mode is activated, the  also becomes .

Reminder:  is the fluid circuit status icon and the gas injection control button. It is displayed in the left corner at the base of the screen.

When the analyzer is in Remote mode, the remote controls reaching the analyzer via the communication protocols (for example: MODBUS) or the digital inputs of the optional Input / Output boards have priority over the HMI controls.

Important remarks:

- The HMI pages open on remote terminals such as computers, smartphones or tablets have the same priority level as the session open on the analyzer screen. The Local / Remote mode is not relevant to them.
- If a fault happens and switches the analyzer to Safety mode , the Remote mode is disabled. If the analyzer was in Remote mode, it automatically switches back to Local Mode. Only local mode operation is allowed. The user must first solve the problem which triggered the Safety mode before reactivating the Remote mode.

Switch the analyzer to Maintenance mode

Double touch / click on  to switch the analyzer to Maintenance mode.  then becomes .

The Maintenance mode is used to give manual access to otherwise locked features, and invalidate measurement data while maintenance operations are being carried out.

When the Maintenance mode is activated, the stored data is tagged with a status bit (see §3.4.2.5).

Double touch / click on  to switch the analyzer back to Measurement mode.

When the analyzer is in Maintenance, the status icon (in the left corner at the base of the screen) is changed to . However, the Alarm status overrides the Maintenance status. Consequently, if the

Maintenance mode is activated while an alarm is triggered, the alarm icon  will remain displayed and the Maintenance icon  will not be displayed. To check whether the Maintenance mode is active, check if the  icon is outlined by a red box in the main Configuration Functions menu.

Switch the analyzer to Standby mode

Double touch / click on  to switch the analyzer to Standby mode.  then becomes .

Standby mode is used to stop the pumps while all the other functions remain in operation.

When the Standby mode is activated, the stored data is tagged with a status bit (see §3.4.2.5).

Double touch / click on  to switch the analyzer back to Measurement mode.

When the analyzer is on Standby, the status icon (in the left corner at the base of the screen) is changed to . However, the Alarm status overrides the Standby status. Consequently, if Standby mode is

activated while an alarm is triggered, the alarm icon  will remain displayed and the Standby icon  will not be displayed. To check whether the Standby mode is active, check if the  icon is outlined by a red box in the main Configuration Functions menu.

Configuration of the optional Input / Output modules

This menu is optional. The  icon is enabled in the main configuration menu when an ESTEL input / output USB module is connected to the analyzer.

Touch or double click on  to enter the menu. This menu is used to test and configure the optional input / output modules connected to the analyzer. There can be at most two such modules. When two modules are connected, the menu comprises two pages. The  pictogram located in the top left corner of the menu identifies the module to which the page match.

PRELIMINARY NOTICE: Some hardware configuration must be done inside the module before proceeding with software configuration. Hardware configuration is required in order to define the following items:

- the address of each module, in order to identify Module n°1 and Module n°2,
- the nature of analog output signals, which can be either 0-1 V, 0-10 V, 4-20 mA or 0-20 mA,
- the ground, which can be common or floating.

Jumpers	State	Role
SW1 to SW4	see photo on the right	Analog output configuration : 0-1 V / 0-10 V / 4-20 mA / 0-20 mA
ST1 to ST4		
ST5		Unused
ST6		Unused
ST7		Address: board n°1
		Address: board n°2
ST8		Common ground
		Floating ground

The board configuration drawing can be displayed by clicking on  and then on or .



Relays and analog outputs manual control page

Touch or click on  to enter this page. When entering the page,  (blue) becomes  (green).

DIGITAL section: provides manual control for the output relays



(yellow): the relay is powered (normally closed)



(green): the relay is idle (normally open)

ANALOG section: provides manual control for the analog outputs. The output signal can be changed either by moving the slider, by entering a numerical value between 0 and 4000 digital counts or by using the MIN / MAX switch and confirming with . This functionality provides an easy way to calibrate signal acquisition systems.



Digital and analog inputs visual display page

Touch or click on  to enter this page. When entering the page,  (blue) becomes  (green).

DIGITAL section: provides a visual display for the digital inputs (remote controls and alarms).

 (yellow): the input is in the high state (the contact is closed across the input pins)

 (green): the input is in the low state (the contact is open across the input pins)

ANALOG section: displays the analog input signals, in mV. The range is 0-2500 mV.

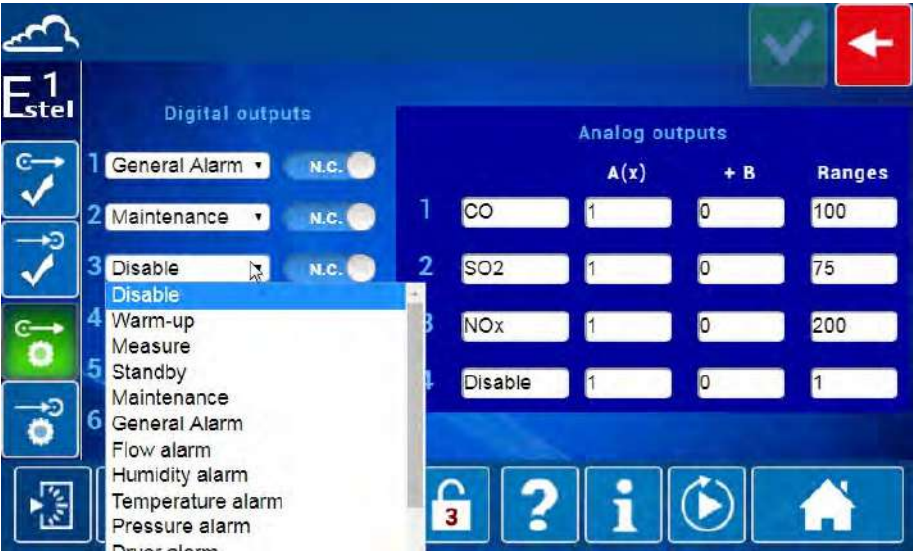


Relays and analog outputs configuration page

Touch or click on  to enter this page. When entering the page,  (blue) becomes  (green).

DIGITAL OUTPUTS section: a function can be allocated to each relay output using the drop downlists. These lists are dynamic: their contents depend on the elements declared as being connected to the analyzer in the Advanced Configuration page (cf § 3.4.2.3) and the Automatic Cycles Configuration page (cf § 3.4.2.4).

The operating mode of each relay can be specified using N.C. (Normally Closed) / N.O. (Normally Open) switches.



The tables below give the full detail of the available relay functions.

States and cycles			
Disabled (default setting)	Zero	Zero cycle	Zero Reference
Warm-up	Span 1	Span cycle 1	Auto cal 1
Measure	Span 2	Span cycle 2	Auto cal 2
Standby	Span 3	Span cycle 3	Auto cal 3
Maintenance	Span 4	Span cycle 4	Auto cal 4
Probe	Span 5	Span cycle 5	Auto cal 5

Solenoid valves control			
SV1	SV3	SV5	SV7
SV2	SV4	SV6	SV8

Alarms		
General alarm	Pressure alarm	Converter oven alarm
Flow alarm	Dryer alarm	Analog. Out of range (=analog output overflow)
Humidity alarm	Probe alarm	Compound X : Overflow Threshold (=compound X out of range)
Temperature alarm	Heated line alarm	

ANALOG OUTPUTS section: provides settings to allocate and calibrate the analog outputs.

For each output:

- Touching / clicking in the first box opens a pop-up window used to allocate a measurement channel to the analog output. All the measurement channels available in the General Configuration menu can be allocated. They are sorted by type, in the same way as in the General Configuration menu.

After selecting the channel, touch / click on  to confirm the selection and close the pop-up window. Touch / click on  to close the pop-up window without changing anything.

- The Range box is used to define the range of values corresponding to the dynamic range of the analog output, in the unit of the allocated channel.
- The A and B calibration parameters must be determined for the signal acquisition system as a whole, using the manual control page. The easiest way to proceed is as below :
 - In the manual control page, click twice on the MIN / MAX switches to set them to MIN, then click on  to set the output analog signals to 0 digital count, which should read as either 0 V, 4 mA or 0 mA depending on the hardware settings.
 - In the configuration page, adjust the B parameters so as to read 0 on the signal acquisition system.
 - In the manual control page, click once on the MIN / MAX switches to set them to MAX, then click on  to set the output analog signals to 4000 digital counts, which should read as either 1 V, 10 V or 20 mA depending on the hardware settings.
 - In the configuration page, adjust the A parameters in order to read the expected maximum signals (1 V, 10 V or 20 mA depending on the hardware settings) on the acquisition system.
 - Repeat the above steps to improve accuracy if necessary.



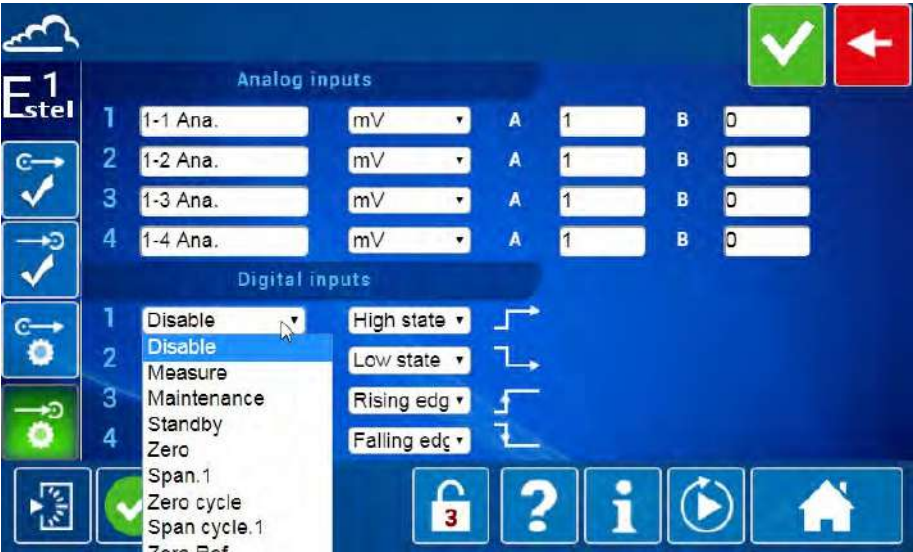
Analog and digital inputs configuration page

Touch or click on  to enter this page. When entering the page,  (blue) becomes  (green).

ANALOG INPUTS section: provide settings to name and calibrate the analog inputs. For each input:

- The first box is used to name the analog input
- The second box is used to choose the unit in which the value will be displayed
- The third and fourth boxes are used to set the calibration factors involved in the conversion between the voltage measured at the input (between 0 and 2500 mV) and the measurement values expressed in the chosen unit. The conversion formula is $y = Ax + B$.

DIGITAL INPUTS section: provides drop down lists to allocate the digital inputs and set their operating modes (*High State, Low State, Rising Edge or Falling Edge*).



The tables below give the full detail of the available digital input functions.

Remote Controls			
Disabled (default setting)	Zero	Zero cycle	Zero Reference
Measure	Span 1	Span cycle 1	Auto cal 1
Maintenance	Span 2	Span cycle 2	Auto cal 2
Standby	Span 3	Span cycle 3	Auto cal 3
Probe	Span 4	Span cycle 4	Auto cal 4
	Span 5	Span cycle 5	Auto cal 5

Alarms		
Dryer alarm	Converter oven alarm	Probe pressure alarm
Probe temperature alarm	Instrument air alarm	Heated line alarm

NOTE : The *High State / Low State* operating modes are good for static and cumulative states such as alarms. *Rising Edge / Falling Edge* operating modes are better suited for instant actions like starting automatic cycles (push button effect).

CALIBRATION

GENERAL

Giống như bất kỳ thiết bị đo lường nào khác, MIR 9000e phải được hiệu chuẩn khi lắp đặt. Điều này liên quan đến việc điều chỉnh Zero bằng cách sử dụng không khí như N₂ hoặc không khí khô, đã được lọc, và sau đó điều chỉnh các hệ số hiệu chuẩn bằng cách sử dụng các chai khí có nồng độ đã biết.

Sau đó, các điểm không và điểm span của khí phải được kiểm tra định kỳ và được điều chỉnh khi cần thiết theo kế hoạch đảm bảo độ chính xác phép đo.

To begin with, it is important to notice that what needs calibrating is rarely the analyzer on its own, but rather part or all of a complete sampling and analysis system. As a result, calibration setups must be different for each type of system.

It is recommended to use one of the predefined system designs described in chapter 2.2. However, if a different system is designed, the following precautions should be taken:

- Pressure at the inlet and outlet of the analyzer must be close to atmospheric pressure. Any pressurized gas delivered to the analyzer should be partially vented to atmospheric pressure prior to entering the analyzer in order to ensure correct pressure.
- If sending calibration gas to the probe, it is important to adjust span cylinder pressures and flows together so that pressure is high enough to get up to the probe, yet flow is low enough when fed directly in the analyzer cabinet.
- Zero gas should be free of any infrared absorbing compound. N₂ cylinders or synthetic air are suitable. Ambient air is suitable too provided that it has been filtered and that CO₂ has been removed using a scrubbing cartridge.
- Dew point should be the same for zero gas, span gas and sample gas. If using a gas cooler, this can be achieved by simply feeding the zero and span gas through the gas cooler. This allows the residual moisture present in the gas cooler to humidify zero and span gas for a long enough time to perform calibration. Ideally, the solenoid valve should be placed between the two stages of a double stage cooler, so that the second stage humidifies the gas, while the first stage protects the valve.
- If using a NO_x converter, it must be part of the subsystem being zeroed and calibrated. This means that zero and span gas must be fed upstream the NO_x converter. Moreover, it is important to take into account any CO and CO₂ produced when using carbon based NO_x converters. Here, flow can be an issue because the concentration of CO and CO₂ produced in the converter depends on flow. It is therefore important to match the flow going through the converter for sample gas measurement, zero mode and span mode. This can be achieved with needle valves as shown in the predefined systems described in chapter 2.2. This is important even if CO and CO₂ are not measured.
- When performing these operations, enough time should be allowed to let the system fully respond to the changes in gas concentrations. Only when measurement values have stabilized should they be used for calibration purposes. As a rule of thumb, for a step change in concentration value, one should wait at least 4 times the amount of time it takes to the system to respond to 90% of the step change.

Zero references, Zero and Span checks and self-calibrations can be performed in different ways:

- Manually, using the manual gas injection controls (see §3.4.1.1),
- Manually, using the automatic cycles controls (see §3.4.1.1),
- Remotely, using a communication protocol,
- Remotely, using digital inputs located on optional I/O modules (see §3.4.2.18).

ZERO REFERENCE

This operation consists of measuring and storing the raw detector signals when no infrared absorbing compound is present in the measurement chamber (see § 2.1). The aim is to determine the reference signals as explained in §2.1, and to adjust the Zero reading of the analyzer.

Zero gas can be either:

- N2 from a cylinder,
- Synthetic air from a cylinder,
- Clean, dry air exempt of any constituents susceptible of affecting the infrared measurement (CO2, oily residues...).

Suitable dew point depends on the installation. If using a gas cooler, any dew point is acceptable as long as the zero gas passes through the cooler. If using a permeation dryer (SEC probe), the instrument air used for the dryer should also be used as a source of zero gas. In that case, a CO2 scrubbing cartridge should be used.

For Zero References, zero gas should be injected directly in the analyzer cabinet, rather than in the sampling probe, although both possibilities exist.

In order to enable zero references, the following parameters must be set up:

- In the Automatic Cycles Configuration page (cf §3.4.2.4):
 - Operating mode, according to system design A, B or C,
 - Solenoid valves setup,
 - Ref. Zero cycle periodicity and duration.

Frequency: depending on the room where the analyzer is located (with or without air conditioning) and local authority requirements. Once a day is normally sufficient.

Duration: typically 600s in total, with a 240s initial purge, 240s integration time, and 120s final purge.

CALIBRATION

Calibration point concentration values are usually defined by local regulations. It is often calculated using the maximum emission values or typical values measured on the installation. If a specific measuring range has been defined, typically the calibration points should lie between 70% and 90% of that range.

The calibration gas cylinder should be certified according to local regulations.

NOTE : When replacing a calibration gas cylinder, it is recommended to compare the span gas analyzer reading before and after the replacement in order to detect any abnormal discrepancy. Agreement between the two readings should be within the cylinder and analyzer tolerances.

If using a gas cooler, calibration gas should pass through the cooler (at least the second stage) so that its dew point is the same as that of the sample gas.

For calibration purposes, span gas injections should be done preferably in the sampling probe, rather than in the analyzer cabinet, although both possibilities exist.

Before calibrating the system, one should perform a successful Zero check or a Zero Reference.

Calibration factors can be adjusted in different ways:

- Either automatically by the analyzer itself,
- By the user,
- Remotely, using a handling system (DAHS).

Automatic self-calibration

In order to enable automatic self-calibration, the following parameters must be setup:

- In the Automatic Cycles Configuration page (cf §3.4.2.4):
 - Operating mode, according to system design A, B or C,
 - Solenoid valves setup, with up to 5 span gas cylinders,
 - Automatic cycles periodicity and duration.
- In the Advances Analyzer Configuration menu, first page (cf §3.4.2.3):
 - In the calib. field, select which solenoid valve drives which span gas,
 - In the Span gas value field, enter the cylinder concentration.

Frequency: usually monthly, although this depends on local authorities and the conditions of use and application. In certain conditions, taking into account the analyzer's low drift, a three-monthly or six-monthly check is sufficient. On the other hand, some authorities require weekly or even daily checks.

Duration: typically 600s in total, with a 240s initial purge, 240s integration time, and 120s final purge.

Manual calibration

The easiest way to perform a manual calibration is to launch the self-calibration cycle by hand, as described in §3.4.1.1.

Alternatively, it is also possible to calibrate the system using the manual gas injection controls and then calculating the new calibration factors using the following formula:

$$K_{(\text{new})} = K_{(\text{oSd})} \times \frac{\text{calibration gas value}}{\text{readout value (without offset)}}$$

Calibration factors are accessible in the first page of the Advanced Analyzer Configuration menu.

Remote calibration using a handling system (DAHS)

The system can be calibrated in different ways when using a remote handling system:

- The DAHS may remotely control the analyzer built-in self-calibration cycles
- The DAHS may remotely control the analyzer built-in calibration check cycles
- The DAHS may remotely control the gas injection

In the last two cases, the DAHS must calculate the new calibration factor by itself, and then either send it to the analyzer or use it internally.

NOTE : The history of all calibration events, including timestamps and calibration factor values, is stored in the analyzer and can be retrieved using the History menu (see §3.4.2.7).

ZERO AND SPAN POINT CHECKS

This operation consists of measuring the system's response to zero gas and then to a gas of known concentration. The aim is to quantify the system's drift over time without changing the analyzer calibration factors.

Frequency: usually monthly, although this depends on the conditions of use and application. In certain conditions, taking into account the analyzer's low drift, a three-monthly or six-monthly check is sufficient. On the other hand, some authorities require weekly or even daily checks.

INITIAL O2 MODULE CALIBRATION

If using a ZrO₂ measurement module for the first time, for example when setting it up or following a replacement, one should use the O₂ Sensor Calibration function  available on the main configuration menu and follow the instructions on the analyzer display.

If using a Paramagnetic measurement module for the first time, proceed as follows:

- Feed N₂ to the analyzer,
- Adjust the Zero trimmer on the rear panel of the analyzer,
- Feed a known concentration (either air or a span gas cylinder) to the analyzer,
- Go to the Diagnostic  page, open the O₂ Probe tab and scroll down to view the O₂ module analog signal value (in mV),
- Adjust the Span trimmer on the rear panel of the analyzer so that the value is between 7800 and 7900 mV.
- Then, run the O₂ Sensor Calibration function as described above.

NOTE : The automatic O₂ sensor calibration function changes the analyzer's *internal* calibration factor. It does not change the *user* calibration factor (named Span. coeff.) available in the Advanced Analyzer Configuration page, unlike the self-calibration functions described earlier in this chapter.

NOTE : The O₂ Sensor Calibration Function uses 20.9% vol. oxygen as its reference, which corresponds to dry air (-10 °C dew point). It is important to keep in mind that ambient air also contains humidity and can have a somewhat lower O₂ concentration. For example: 20.7% vol. if the air contains 1% vol. humidity (7°C dew point); 20.6% vol for 1.5% vol humidity (13°C dew point); 20.5% vol. for 2% vol. humidity (17°C dew point).

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